

Final Project

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The Effects of Internal Population Migration in American Politics

It is no secret that America's political landscape is in a constant state of flux, and the lines of its strategic map are redrawn every election. News media generally focuses on the divisions of the two parties, often irresponsibly leaning into party polarity as if covering rival sports teams, but other important factors are in play as well. Demographics is one. In 2018, the Brookings Institute demonstrated how—because of natural demographic changes in the size and composition of the population that result from births, deaths, and aging—“minorities will be the source of all of the growth in the nation's youth and working age population, most of the growth in its voters, and much of the growth in its consumers and tax base as far into the future as we can see” (Frey (2018)). Another widely reported factor is immigration: according to the Pew Research Center, over “40 million people living in the U.S. were born in another country” (Budiman (2020)). One comparatively understudied area of politics, however, does not depend on the growth of the voter base, but in how citizens are distributed. Because of the narrow margins of victory in battleground states, focus on American internal migration is vital in predicting election outcomes, and analytical tools such as R are especially apt to glean new truths from the data.

Research Questions

The extent of internal migration is measured by the Census Bureau in the American Community Survey. Of the 127,544,730 American households in 2021, 34,784,732 have changed residents since 2019. The 2021 American Community Survey also shows that approximately 7,000,000 people—2.4% of the United States population—moved from one state to another. Moreover, the migration is heterogeneous, and much is from Democratic dominated metropolitan areas, part of what Derek Thompson calls a “blue urban exodus” (Thompson (2019)). In fact, of the “13 percent of Americans [who] moved” from 2017-2018, 25 percent were from the 18-24 group, versus “6 percent of individuals age 65 and over” (Frost (2020)). Likewise, low-income Americans had higher residential mobility than their wealthier counterparts.

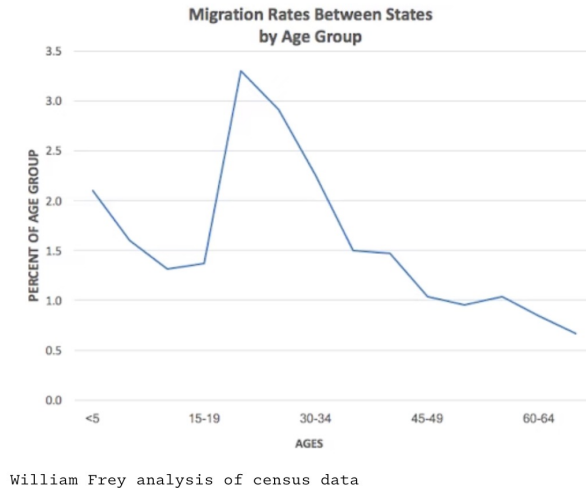


Figure 1: Line graph showing how much more mobile are young adults

Americans aged 20 to 40 are three times more likely to move between states than their elderly counterparts from 50 to 70. Though this is not by itself a large number of people, it is politically significant, because four states (Pennsylvania, Wisconsin, Arizona, and Georgia) in the 2020 election were decided within two percent.

Moreover, the effects of large numbers of citizens in motion provide an insight into the voting districts that extend far beyond the vote counting. Such trends can be an outsized indicator at how attractive a district is to a prospective citizen of a given background and may provide clues that can help answer numerous questions across America’s political map.

For example, Florida has long been a magnet for citizens of other states, especially from the North, where it drew large numbers of Democratic voters to offset a dominant base of Cuban immigrants who have historically voted Republican. For years, political analysts have considered Florida to be a bellwether or swing state, but today such claims are disputable, because Democrats have won only 3 presidential elections in Florida since 1976, and its governor, Ron DeSantis, is acknowledged to be a favorite for the Republican nomination. Does domestic migration provide any clues to the state’s political future?

Democrats in the 2020 Georgia election, on the other hand, broke a string of six Republican victories, and its contested “purple” status was confirmed by the 2022 senatorial elections. To what extent is domestic migration a factor in the political landscape in Metro-Atlanta’s counties (Cobb, Gwinnett, Henry, Fayette, Rockdale, Newton, Douglas), and what does the future hold?

Maricopa County in Arizona is another area of focus. To what degree is domestic migration a factor in Arizona’s politics? Has it changed the election outcomes?

Republicans in Texas, which will have 40 electoral votes in 2024 (two more than the 2020 election), won the state by over 5 percentage points, however, the difference was 9 percent in 2016, and nearly 16 percent in 2012. Tarrant County, home of Ft Worth, flipped to Democratic in 2020, while others like Denton and Collin also trended left. Is voter migration a significant factor in these changes?

Oakland County, Michigan and Hamilton County (of Cincinnati, Ohio) are other areas undergoing change. Oakland County used to be a solid red county, but that is no longer true. Hamilton County is also trending blue, even though Ohio, by and large, remains very conservative. To what extent is voter migration a factor that causes counties like Oakland, Hamilton, or Delaware, Montgomery, and Chester counties in Pennsylvania to buck the trend of their states? Are these merely urban to suburban migrations or are there other forces at work?

Finally, overall, what do these trends mean for the political parties and their candidates? Are the trends sustainable or a momentary aberration?

Approach

The most important decision for this project was the abstraction of a voter. While opinion polls are ubiquitous during election season, I did not want to rely on self-reporting. The expectation of voter privacy is a fundamental right of our democracy, so how an individual votes can only be guessed from wider outcomes. Moreover, although states make voter rolls public, access is not free, nor is it for any interested party. Also, there is no uniform data base. Instead, each state makes its own rules, and politicians and strategists pay for access to such bodies as the National Conference of State Legislatures. It is also expensive: Access to Alabama's alone through NCSL.org would cost over \$40000.

Instead, I will define the definition of a potential voter by the political bias of their source residency. To denote the valence of partisan and ideological identities this project uses the results of recent presidential elections and assigns crude red-vs-blue weights for both the districts studied and those that contribute to their change. A vote for a candidate is assumed to define the characteristics of the voters of the precinct. The sum of the inflow and outflow of voter entities from source locations into a target of study, collected on an annual basis, is used to produce a model and predict future results, which is further evaluated to determine the model's efficacy. This can further be refined to achieve a better model.

For example, I live in Florida's 7th Congressional District, which in 2020 (before redistricting) contained portions of two counties (Seminole and Orange) and twenty-four different zip codes:

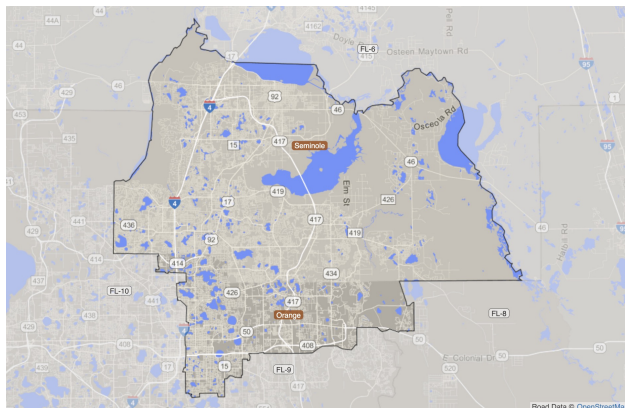


Figure 2: Map showing 7th Congressional District of Florida by county

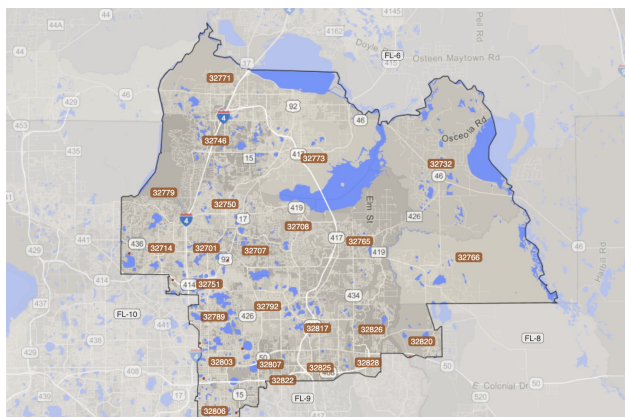


Figure 3: Map showing 7th Congressional District of Florida by zip

The 7th Congressional district is part of the I4-Corridor, one of the last areas of Florida that are not seen as deep red or deep blue. Candidates for high office always visit this region of 7 counties and spend money in its

media market to bombard potential voters with advertisements. In 2020, 54.5% of voters in the 7th chose the eventual winner, Joe Biden, versus 44.15 who opted for then-President Trump.

Like many states, Florida tracks voter participation by county rather than by precinct. In the general election, 77% of eligible voters cast their ballot, the highest percentage since 1992. The distinction between precinct-level tracking of votes and county-level tracking of voter participation is representative of the challenging nature of location in residential mobility. The holy grail of relocation data is the National Change of Address Link, which is maintained by the post office who sells the data freely provided by its customers (unless they opt out) to a multitude of commercial vendors and nonprofits. The data is specific to an address—using 9-digit zip codes. Unfortunately, access to this data requires a subscription at a cost beyond the project budget.

Another alternative is provided by the Internal Revenue Service, but it has two serious deficiencies: The intersection between voters and taxpayers is inexact. Some non-taxpayers are voters, while some taxpayers are unregistered or forbidden to vote. Moreover, the IRS address-change data is county-specific, rather than zip code specific, which means new inaccuracies are introduced. Nevertheless, the data is quite useful for a proof-of-concept, and several pathways exist to use intermediate data sets to improve the results. Best of all, the IRS data is free and easily available.

Data: Sources, Importing, and Cleaning

Two choices were available for the election outcomes. One, a DOLT dataset I used in the first step of the project, has the advantage of being precinct-based, which could be used in conjunction with the zip-code post office records for migration. However, the data set only included the 2016 and 2020 elections.

County-based election results were available in a tab-delimited data base from DataVerse:

```
#https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/VOQCHQ
```

Before the data could be used, a number of steps were required.

Initial steps for transforming election results: * Change tab-delimited file to comma-delimited file. * Use rio package “import” to load csv file. * Give columns names * Remove data that is unnecessary: + mode (of voting) + date of data-set (Spring 2022) + office sought (President) + state name (redundant with state-po abbreviation)

```
# Use the data from:
#https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/VOQCHQ
# It is more complete and reliable than the Dolt data base from last week.

# It needs to be changed from a tab separated file to a csv first.
writeLines(gsub("\t", ",", readLines("data/countypres_2000-2020.tab")),
           "data/countypres_2000-2020.csv")

# load it in.
election_results_df <- rio::import("data/countypres_2000-2020.csv")

# give the columns some names.
names(election_results_df) <- c("year", "state", "state_po", "county_name",
                               "county_fips", "office", "candidate", "party",
                               "candidate_votes", "total_votes",
                               "data_set_date", "mode")

# drop some columns we don't need now. (mode, data_set_date, office, state)
election_results_df <- election_results_df[,-12]
election_results_df <- election_results_df[,-11]
```

```
election_results_df <- election_results_df[,-6]
election_results_df <- election_results_df[,-2]
```

Another important dataset are the migrations of citizens between counties. This is a set of 18 files, two for each year from 2011 through 2019. They are csv files, one for inflow and the other for outflow. The IRS data that describes inflow and outflow of counties is described here:

<https://www.irs.gov/statistics/soi-tax-stats-migration-data-2019-2020>

<https://www.irs.gov/pub/irs-soi/1920inpublicmigdoc.pdf>

The sourcing of the data is explained in the documentation: “Migration data are based on the population of Forms 1040 that were filed and processed by the IRS during calendar years 2019 and 2020. In addition, tax year 2019 paper returns were included up until mid-July of 2021 because of processing delays due to the COVID-19 pandemic.”

I use the IRS’s county-specific data instead of the state-specific data. It has a few more fields. Here is a complete list of the columns:

Note: For both inflow and outflow, Y1=Year One, Y2=Year Two. For a given transaction, the county just changes columns in the datasets.

1. Y1_STATEFIPS: State FIPS Code of Origin from Year 1. Numbered alphabetically. Alabama = 1, Wyoming = 56. 57-59 summary records, 96-98 likewise.
2. Y1_COUNTYFIPS County FIPS Code of Origin from Year 1. (state-total=0, 001-840 are the potential values)
3. Y2_STATEFIPS State FIPS Code of Destination from Year 2
4. Y2_COUNTYFIPS County FIPS code of Destination from Year 2
5. Y2_STATE State Abbreviation or Postal Code of Destination from Year 2. “AL” to “WY”. “FR” means “foreign”. “SS” means “same state”.
6. Y2_COUNTYNAME County Name of Destination from Year 2. For summary fields it has special values: [State County Name] Total Migration – US and Foreign [State County Name] Total Migration – US [State County Name] Total Migration – Same State [State County Name] Total Migration – Different State [State County Name] Total Migration – Foreign For non-migrant records, the name will take the following format: [State County Name] Non-migrants For the foreign records, the name will take the following format: Foreign – Overseas Foreign – Puerto Rico Foreign – APO/FPO ZIPs Foreign – US Virgin Islands
7. N1 Number of returns <- It was actually used for tax purposes.
8. N2 Number of individuals <- Individuals is more useful for us.
9. AGI Adjusted Gross Income (AGI) <- We do not use this field.

To find trends of migration, which are vital to building a model, I import each year of the data. It is self-reported to the IRS and (as previously specified) is a count of taxpayers, not voters. Even so, as an indicator of trends of migration, it is still valid, because taxpayers are a population subset too and we are working with percentages and not absolutes most of the time.

These datasets have a number of peculiarities that must be addressed, but first they need to be imported. I also used the “import” function of the rio package, a Swiss army-knife tool that guesses data format from the file extension (in this case comma-delimited, “csv”).

```
# County inflow and outflow files describe the migration of each county.
# This is a set of 18 files. Make the filenames first.
inflow_file_names <- paste0('data/countyinflow', 11:19, 12:20, '.csv')
outflow_file_names <- paste0('data/countyoutflow', 11:19, 12:20, '.csv')
# lists for the data
inflow_data <- list()
outflow_data <- list()
# import the data from the files by calling import for each name.
```

```
inflow_data <- lapply(inflow_file_names, rio::import)
outflow_data <- lapply(outflow_file_names, rio::import)
```

To link the migration data with the election results, I needed a common key that denoted each county. The “FIPS” is a federal identifier, but its format varies between these databases. This code makes the FIPS of the migration data conform to the format of the election results. It also removes special summary records as well as foreign migration.

```
# Tweak the county id, "Fips", to make it the same across all data.
# Functions for padding with zeros.
# Get rightmost n characters of a string s
substrRight <- function(s, n){
  substr(s, nchar(s)-n+1, nchar(s))
}
# With padded zeros.
substrRightPaddedWithLeft <- function(s, n, z) {
  padded_s <- str_pad(s, n, side='left', pad=z)
  substrRight(padded_s, n)
}

# transform fips for all the county migration data.
for (i in 1:length(inflow_data)) {
  county_inflow_df <- inflow_data[[i]]
  county_outflow_df <- outflow_data[[i]]
  # Exclude special designators (for totals, foreign immigration etc)
  # remove statefips 57-59 summary records, 96-98 likewise
  # The only other data to possibly exclude are when the county and state
  # are the same as the inflow (or outflow), which means they stayed in place.
  # Keeping it for now.
  county_inflow_df <- county_inflow_df[county_inflow_df$y1_statefips<57,]
  county_outflow_df <- county_outflow_df[county_outflow_df$y2_statefips<57,]
  # Keys for all inflow counties
  county_inflow_df$y1_combined_fips <-
    paste(county_inflow_df$y1_statefips,
          substrRightPaddedWithLeft(county_inflow_df$y1_countyfips, 3, "0"),
          sep = "")
  county_inflow_df$y2_combined_fips <-
    paste(county_inflow_df$y2_statefips,
          substrRightPaddedWithLeft(county_inflow_df$y2_countyfips, 3, "0"),
          sep = "")
  # Keys for all outflow counties
  county_outflow_df$y1_combined_fips <-
    paste(county_outflow_df$y1_statefips,
          substrRightPaddedWithLeft(county_outflow_df$y1_countyfips, 3, "0"),
          sep = "")
  county_outflow_df$y2_combined_fips <-
    paste(county_outflow_df$y2_statefips,
          substrRightPaddedWithLeft(county_outflow_df$y2_countyfips, 3, "0"),
          sep = "")
  inflow_data[[i]] <- county_inflow_df
  outflow_data[[i]] <- county_outflow_df
}

# we do not need these anymore
```

```

county_inflow_df<-NULL
county_outflow_df<-NULL

# guarantee the inflow and outflow data has no duplicate records.
for (migration_year in 2011:2019) {
  inflow_data[[migration_year - 2010]]<-unique(inflow_data[[migration_year - 2010]])
  outflow_data[[migration_year - 2010]]<-unique(outflow_data[[migration_year - 2010]])
}

```

Construction of the County Valence Calendar

My next step is to build the County Valence Calendar data frame that denotes the political valence of each county for each election. Essentially, the presidential election results are used to assign the valence from 0 (red=Republican) to 1 (blue=Democrat). It would also be nice to add voter turnout here but as of now, I have not discovered any good national data base with turnout data. Each state has its own method of tracking turnout.

This is the format of the County Valence Calendar data frame:

1. combined_fips: The county identifier.
2. county_name: Keep the name for interesting reports.
3. state_po: Used to select by states. 4-8. democratic_votes_2004, democratic_votes_2008 ... democratic_votes_2020 9-13. republican_votes_2004, republican_votes_2008 ... republican_votes_2020 14-18. valence_2004, valence_2008... valence_2020

First field is a fips, copied from the election_results file.

```

county_fips <- unique(election_results_df$county_fips)
county_valence_calendar_df <- data.frame(county_fips)

```

To compute the valence the data is divided into segments for each party and each election.

```

democratic_2004_votes_df <- election_results_df[
  election_results_df$party=="DEMOCRAT" & election_results_df$year==2004,]
republican_2004_votes_df <- election_results_df[
  election_results_df$party=="REPUBLICAN" & election_results_df$year==2004,]
democratic_2008_votes_df <- election_results_df[
  election_results_df$party=="DEMOCRAT" & election_results_df$year==2008,]
republican_2008_votes_df <- election_results_df[
  election_results_df$party=="REPUBLICAN" & election_results_df$year==2008,]
democratic_2012_votes_df <- election_results_df[
  election_results_df$party=="DEMOCRAT" & election_results_df$year==2012,]
republican_2012_votes_df <- election_results_df[
  election_results_df$party=="REPUBLICAN" & election_results_df$year==2012,]
democratic_2016_votes_df <- election_results_df[
  election_results_df$party=="DEMOCRAT" & election_results_df$year==2016,]
republican_2016_votes_df <- election_results_df[
  election_results_df$party=="REPUBLICAN" & election_results_df$year==2016,]
democratic_2020_votes_df <- election_results_df[
  election_results_df$party=="DEMOCRAT" & election_results_df$year==2020,]
republican_2020_votes_df <- election_results_df[
  election_results_df$party=="REPUBLICAN" & election_results_df$year==2020,]

```

Add up the candidate votes for 2020, because the source data set has multiple entries, one for each type of vote: mail-in, provisorial, and so forth. None of these distinctions are relevant for this project.

```
# consolidate multiple entries for 2020 depending on the type of ballot cast.
```

```
democratic_2020_votes_df <- democratic_2020_votes_df %>%  
  group_by(county_fips) %>%  
  mutate(candidate_votes = sum(candidate_votes)) %>%  
  ungroup()
```

```
republican_2020_votes_df <- republican_2020_votes_df %>%  
  group_by(county_fips) %>%  
  mutate(candidate_votes = sum(candidate_votes)) %>%  
  ungroup()
```

```
# The 2020 results have duplicate rows for different modes.
```

```
# Get rid of those now.
```

```
democratic_2020_votes_df <- distinct(democratic_2020_votes_df)  
republican_2020_votes_df <- distinct(republican_2020_votes_df)
```

Recombine everything into one registry, each record representing one county.

```
county_valence_calendar_df <- merge(county_valence_calendar_df,  
  democratic_2004_votes_df[c("county_fips",  
                             "county_name",  
                             "state_po",  
                             "candidate_votes")],  
  by = "county_fips",  
  all.candidate_votes = TRUE)  
county_valence_calendar_df <- rename(county_valence_calendar_df,  
  democratic_2004_votes = candidate_votes)  
# merge next results by fips  
county_valence_calendar_df <- merge(county_valence_calendar_df,  
  democratic_2008_votes_df[c("county_fips",  
                             "candidate_votes")],  
  by = "county_fips",  
  all.candidate_votes = TRUE)  
county_valence_calendar_df <- rename(county_valence_calendar_df,  
  democratic_2008_votes = candidate_votes)  
county_valence_calendar_df <- merge(county_valence_calendar_df,  
  democratic_2012_votes_df[c("county_fips",  
                             "candidate_votes")],  
  by = "county_fips",  
  all.candidate_votes = TRUE)  
county_valence_calendar_df <- rename(county_valence_calendar_df,  
  democratic_2012_votes = candidate_votes)  
county_valence_calendar_df <- merge(county_valence_calendar_df,  
  democratic_2016_votes_df[c("county_fips",  
                             "candidate_votes")],  
  by = "county_fips",  
  all.candidate_votes = TRUE)  
county_valence_calendar_df <- rename(county_valence_calendar_df,  
  democratic_2016_votes = candidate_votes)  
county_valence_calendar_df <- merge(county_valence_calendar_df,  
  democratic_2020_votes_df[c("county_fips",  
                             "candidate_votes")],  
  by = "county_fips",
```

```

all.candidate_votes = TRUE)
county_valence_calendar_df <- rename(county_valence_calendar_df,
  democratic_2020_votes = candidate_votes)
county_valence_calendar_df <- merge(county_valence_calendar_df,
  republican_2004_votes_df[c("county_fips",
    "candidate_votes")],
  by = "county_fips",
  all.candidate_votes = TRUE)
county_valence_calendar_df <- rename(county_valence_calendar_df,
  republican_2004_votes = candidate_votes)
# merge next results by fips
county_valence_calendar_df <- merge(county_valence_calendar_df,
  republican_2008_votes_df[c("county_fips",
    "candidate_votes")],
  by = "county_fips",
  all.candidate_votes = TRUE)
county_valence_calendar_df <- rename(county_valence_calendar_df,
  republican_2008_votes = candidate_votes)
county_valence_calendar_df <- merge(county_valence_calendar_df,
  republican_2012_votes_df[c("county_fips",
    "candidate_votes")],
  by = "county_fips",
  all.candidate_votes = TRUE)
county_valence_calendar_df <- rename(county_valence_calendar_df,
  republican_2012_votes = candidate_votes)
county_valence_calendar_df <- merge(county_valence_calendar_df,
  republican_2016_votes_df[c("county_fips",
    "candidate_votes")],
  by = "county_fips",
  all.candidate_votes = TRUE)
county_valence_calendar_df <- rename(county_valence_calendar_df,
  republican_2016_votes = candidate_votes)
county_valence_calendar_df <- merge(county_valence_calendar_df,
  republican_2020_votes_df[c("county_fips",
    "candidate_votes")],
  by = "county_fips",
  all.candidate_votes = TRUE)
county_valence_calendar_df <- rename(county_valence_calendar_df,
  republican_2020_votes = candidate_votes)
# executing these merges somehow led to empty row slots.
county_valence_calendar_df <- na.omit(county_valence_calendar_df)
county_valence_calendar_df <- unique(county_valence_calendar_df)

```

Now the valences can easily be determined and the valence calendar will be complete.

```

county_valence_calendar_df <- county_valence_calendar_df %>%
  mutate(valence_2004 = democratic_2004_votes/
    (democratic_2004_votes+republican_2004_votes)) %>%
  mutate(valence_2008 = democratic_2008_votes/
    (democratic_2008_votes+republican_2008_votes)) %>%
  mutate(valence_2012 = democratic_2012_votes/
    (democratic_2012_votes+republican_2012_votes)) %>%
  mutate(valence_2016 = democratic_2016_votes/
    (democratic_2016_votes+republican_2016_votes)) %>%

```

```
mutate(valence_2020 = democratic_2020_votes/
      (democratic_2020_votes+republican_2020_votes))
```

A peek at this data, which is used as a lookup table for the processing of records of the county migration data, is informative. First, I will show the results of Orange County and Seminole County of my congressional district:

```
demo_states_subset <- c("FL")
demo_counties_subset <- c("SEMINOLE", "ORANGE")
demo_state_test <- county_valence_calendar_df$state_po %in% demo_states_subset
demo_county_test <- county_valence_calendar_df$county_name %in% demo_counties_subset

# Subset results_df to only include the rows where state_po
# is in the battleground_states array
demo_calendar_df <- county_valence_calendar_df[demo_county_test & demo_state_test,]

kable(demo_calendar_df[1:length(demo_counties_subset), c(1,2,3,7,8,12,13,17,18)],
      col.names = c("fips", "county", "state", "D2016", "D2020",
                    "R2016", "R2020", "V2016", "V2020"),
      row.names = FALSE)
```

fips	county	state	D2016	D2020	R2016	R2020	V2016	V2020
12095	ORANGE	FL	329894	395014	195216	245398	0.6282379	0.6168123
12117	SEMINOLE	FL	105914	132528	109443	125241	0.4918066	0.5141347

The Seminole result of 0.514 shows a slight lean towards blue, which is a fair assessment for 2020.

Here is another example across a number of counties (as well as some synonymous interlopers!) discussed in the research questions.

```
demo_states_subset <- c("AZ", "GA", "MI", "OH", "PA", "TX")
demo_counties_subset <- c("COBB", "DOUGLASS", "FAYETTE", "GWINETT",
                        "HENRY", "NEWTON", "ROCKDALE",
                        "MARICOPA", "COLLIN", "DENTON", "TARRANT",
                        "CHESTER", "DELAWARE", "HAMILTON", "MONTGOMERY")
demo_state_test <- county_valence_calendar_df$state_po %in% demo_states_subset
demo_county_test <- county_valence_calendar_df$county_name %in% demo_counties_subset

# Subset results_df to only include the rows where state_po
# is in the battleground_states array
demo_calendar_df <- county_valence_calendar_df[demo_county_test & demo_state_test,]
demo_calendar_df <- demo_calendar_df %>%
  mutate(location = paste(demo_calendar_df$county_name, demo_calendar_df$state_po, sep=","))

kable(demo_calendar_df[1:length(demo_counties_subset), c(1,19,7,8,12,13,17,18)],
      col.names = c("fips", "location", "D2016", "D2020",
                    "R2016", "R2020", "V2016", "V2020"),
      row.names = FALSE)
```

fips	location	D2016	D2020	R2016	R2020	V2016	V2020
4013	MARICOPA, AZ	702907	1040774	747361	995665	0.4846739	0.5110755
13067	COBB, GA	160121	221846	152912	165459	0.5115148	0.5727941
13113	FAYETTE, GA	23284	33065	35048	37952	0.3991634	0.4655927

fips	location	D2016	D2020	R2016	R2020	V2016	V2020
13151	HENRY, GA	50057	73276	45724	48187	0.5226193	0.6032784
13209	MONTGOMERY, GA	847	979	2670	2960	0.2408303	0.2485402
13217	NEWTON, GA	21943	29794	20913	23869	0.5120170	0.5552056
13247	ROCKDALE, GA	23255	31244	13478	13012	0.6330820	0.7059834
39041	DELAWARE, OH	40872	57735	57568	66356	0.4151971	0.4652634
39047	FAYETTE, OH	2739	2975	7995	9473	0.2551705	0.2389942
39061	HAMILTON, OH	215719	246266	173665	177886	0.5540007	0.5806079
39069	HENRY, OH	3756	4062	9301	10479	0.2876618	0.2793481
39113	MONTGOMERY, OH	122016	135064	123909	129034	0.4961513	0.5114162
42029	CHESTER, PA	141682	182372	116114	128565	0.5495896	0.5865240
42045	DELAWARE, PA	177402	206423	110667	118532	0.6158316	0.6352356
42051	FAYETTE, PA	17946	20444	34590	41227	0.3415943	0.3315010

The final three columns show the political valence (0=Republican, 1=Democratic) in the last three elections. The 2004 and 2008 data is also stored in the data frame, so longer trends can also be computed. The 2000 data is also available but not used in this assignment, because there is no IRS inflow/outflow available before 2009.

As a side note, county also means county-equivalent, such as the District of Columbia, Alaska boroughs and census areas, Louisiana parishes, and the 41 independent cities: Carson City, Nevada, St. Louis, Missouri, Baltimore, Maryland, and 38 in Virginia. There are also the 8 districts of Connecticut that are not county governments but still separate in the databases.

This is not quite right though, because it contains entries for counties in different states with the same name: Montgomery County in Georgia, for example. For purposes of illustration, though, it is fine. This is only a lookup table, and it must work for Montgomery County in Georgia and Fayette County in Ohio too!

These valences are a crude estimate of the county's political leaning. Similarly, the inflow citizens' political leanings can also be assessed and contrasted with the current makeup of the county's voting base. Projecting this data forward provides an indication of how residential mobility will change the precincts of the county. Ideally, turnout information could also be used in the calendar, which would better assess the impact of voters who originate from the locations. Until such data is included, turnout is assumed to be uniform, which it is not. This is one serious limitation of the project that should be addressed.

As previously stated, residential mobility is only one factor in the changing political spectrum, so this tool can only provide a partial explanation to the changing political landscape for each of the places discussed in the section of questions. Nevertheless, the insight it provides is vital, because other sociological factors drive these changes, for example economic factors, a changing employment base, or evolved educational shift. Moreover, like most big data problems, the value of the solution is larger when viewed not on the impact of a singular precinct, but rather on the macro effects of all these areas together.

Now that the data has been loaded and converted, the next step is to build a target data frame, Results or results_df, from the county migration data described earlier and the County Valence Calendar just constructed. The format of these files are the same, a set of 21 columns:

1. The FIPS, our unique key described before.
2. The county name.
3. The state abbreviation.
4. Nine columns of valence change, essentially the sum of all the county immigrants and county emigrants multiplied by their valence of source or destination county.
5. Nine columns of the count of citizens who remain who are the population base.

```

results_df <- data.frame(county_fips)
results_df <- results_df %>% merge(county_valence_calendar_df[c("county_fips",
                                                             "county_name",
                                                             "state_po")],
                                by = "county_fips")

# Areas considered battlegrounds in the 2020 election were
# Arizona, Florida, Georgia, Iowa, Maine, Michigan, Minnesota, Nebraska,
# Nevada, New Hampshire, North Carolina, Ohio, Pennsylvania, Texas and Wisconsin

battleground_states <- c("AZ", "FL", "GA", "IA", "ME", "MI", "MN", "NE",
                        "NV", "NH", "NC", "OH", "PA", "TX", "WI")

# battleground_states <- c("AZ", "FL", "GA", "MI",
#                          "NV", "OH", "PA", "TX")

# Create a logical vector indicating whether each row's state_po is
# in the battleground_states array
matches_battleground <- results_df$state_po %in% battleground_states

# Subset results_df to only include the rows where state_po is
# in the battleground_states array
battleground_df <- results_df[matches_battleground,]
# fips_to_test has all counties in battleground states.
fips_to_test <- battleground_df$county_fips

```

New vector fields are created for battleground_df whose data is produced as the counties are processed. Valence changes is the net political effect change for the citizens who move into and out of the counties. For example, in reference to the previous data, suppose 200 people move from Henry County, Georgia to Newton County, Georgia in 2017, a year for which the valence of the 2016 election is used. These 200 people cause a valence change of $200 * 0.5226193$ or +104.5 to Newton County and a valence change of $-200 * 0.5120170$ or -102 to Henry County.

```

# 9 columns of one row for each county.
valence_changes <- array(0, dim = c(length(fips_to_test), 9))
static_citizens <- array(0, dim = c(length(fips_to_test), 14))

```

Examine each county and its migrations to build the results for the selected counties.

```

# 9 columns of one row for each county.
valence_changes <- array(0, dim = c(length(fips_to_test), 9))
static_citizens <- array(0, dim = c(length(fips_to_test), 14))

for (migration_year in 2011:2019) {
  inflow_data[[migration_year - 2010]] <- unique(inflow_data[[migration_year - 2010]])
  outflow_data[[migration_year - 2010]] <- unique(outflow_data[[migration_year - 2010]])
}

for (fip_ix in 1:length(fips_to_test)) {
  key_to_match <- fips_to_test[fip_ix]
  for (migration_year in 2011:2019) {
    # get all records for this county
    matching_immigrant_rows_df <- filter(inflow_data[[migration_year - 2010]],
                                         inflow_data[[migration_year - 2010]]$y2_combined_fips == key_to_match)
    matching_emigrant_rows_df <- filter(outflow_data[[migration_year - 2010]],

```

```

    outflow_data[[migration_year - 2010]]$y1_combined_fips == key_to_match)
# remove the ones who stayed in place
static_citizen_df <-
  matching_emigrant_rows_df[matching_emigrant_rows_df$y2_combined_fips ==
    key_to_match,]
if (nrow(static_citizen_df)==0) {
  if (migration_year == 2011) {
    # a population of 1 is the minimum for the log to work
    static_citizens[fip_ix, 1]<-1
  }
  else
    static_citizens[fip_ix, migration_year - 2010]<-
      static_citizens[fip_ix, migration_year - 2010 - 1]
}
else
  static_citizens[fip_ix, migration_year - 2010]<-static_citizen_df$n2
matching_immigrant_rows_df <-
  matching_immigrant_rows_df[-(matching_immigrant_rows_df$y1_combined_fips
    == key_to_match),]
matching_emigrant_rows_df <-
  matching_emigrant_rows_df[-(matching_emigrant_rows_df$y2_combined_fips
    == key_to_match),]

# sum of people (we aren't counting this now, but it might be useful later)
new_immigrants <- sum(matching_immigrant_rows_df$n2)
new_emigrants <- sum(matching_emigrant_rows_df$n2)
# add valence amounts for all immigrant and emigrant rows.
# first we need to make sure the name of the fields we merge match.
matching_immigrant_rows_df<-
  rename(matching_immigrant_rows_df, county_fips = y1_combined_fips)
matching_emigrant_rows_df<-
  rename(matching_emigrant_rows_df, county_fips = y2_combined_fips)
# do the merge to transfer the political valence of counties into immigrant rows
matching_immigrant_rows_df <-
  merge(matching_immigrant_rows_df,
        county_valence_calendar_df[c("county_fips",
                                      "valence_2004",
                                      "valence_2008",
                                      "valence_2012",
                                      "valence_2016",
                                      "valence_2020")],
        by = "county_fips")
matching_emigrant_rows_df <- merge(matching_emigrant_rows_df,
        county_valence_calendar_df[c("county_fips",
                                      "valence_2004",
                                      "valence_2008",
                                      "valence_2012",
                                      "valence_2016",
                                      "valence_2020")],
        by = "county_fips")
# we reuse the fields emigrant_change and immigrant_change to calculate.
# they are just the number of people "n2" times the valence of their county.
if (migration_year >= 2019) {

```

```

    matching_emigrant_rows_df <- matching_emigrant_rows_df %>%
      mutate(emigrant_change = round(n2 * valence_2020),0)
    matching_immigrant_rows_df <- matching_immigrant_rows_df %>%
      mutate(immigrant_change = round(n2 * valence_2020),0)
  }
  else if (migration_year >= 2015) {
    matching_emigrant_rows_df <- matching_emigrant_rows_df %>%
      mutate(emigrant_change = round(n2 * valence_2016),0)
    matching_immigrant_rows_df <- matching_immigrant_rows_df %>%
      mutate(immigrant_change = round(n2 * valence_2016),0)
  } else if (migration_year >= 2011) {
    matching_emigrant_rows_df <- matching_emigrant_rows_df %>%
      mutate(emigrant_change = round(n2 * valence_2012),0)
    matching_immigrant_rows_df <- matching_immigrant_rows_df %>%
      mutate(immigrant_change = round(n2 * valence_2012),0)
  } else {
    matching_emigrant_rows_df <- matching_emigrant_rows_df %>%
      mutate(emigrant_change = round(n2 * valence_2008),0)
    matching_immigrant_rows_df <- matching_immigrant_rows_df %>%
      mutate(immigrant_change = round(n2 * valence_2008),0)
  }
  # Add up the changes. We do it separately because later it might be fun
  # to see how each of these vary.
  immigrant_change_sum <- sum(matching_immigrant_rows_df$immigrant_change)
  emigrant_change_sum <- sum(matching_emigrant_rows_df$emigrant_change)
  # Calculate the valence change for this year's citizens.
  valence_changes[fip_ix, migration_year - 2010] <- immigrant_change_sum -
    emigrant_change_sum
}

# Fit a linear regression model to the data
years <- 2011:2019
populations <- static_citizens[fip_ix, 1:9]
population_model <- lm(log(populations) ~ years)
growth_rate <- exp(coef(population_model)[2])

# Estimate the population in future years
future_years <- 2020:2025
future_populations <- numeric(length(future_years))
future_populations[1] <- populations[length(populations)]
for (future_year in 2:length(future_years)) {
  future_populations[future_year] <-
    round(future_populations[future_year-1] * growth_rate, digits = 0)
}
for (migration_year in 2020:2024) {
  static_citizens[fip_ix, migration_year - 2010] <-
    future_populations[migration_year-2019+1]
}
}

```

Finally, add the vectors to the results data frame.

```

battleground_df[, c("chg_2011", "chg_2012", "chg_2013",
  "chg_2014", "chg_2015", "chg_2016",
  "chg_2017", "chg_2018", "chg_2019")] <- valence_changes

```

```
battleground_df[, c("static_2011", "static_2012", "static_2013",
                    "static_2014", "static_2015", "static_2016",
                    "static_2017", "static_2018", "static_2019",
                    "static_2020", "static_2021", "static_2022",
                    "static_2023", "static_2024")] <- static_citizens
```

For now, this is the data's final form.

Data Analysis and Predictions

Now that the data is computed, I can use it to analyze its mean and project future outcomes. First i slice and dice the data into a table with three values: x, y, and n, where “x” is the year, “n” is the county population, and “y” is the valence change.

```
# First, create a new data frame df_long with columns "year" and "y",
# where "year" has the values "chg_2011", "chg_2012", "chg_2013", etc,
# and "y" has the corresponding values for each year
```

```
df_long <- gather(battleground_df, key = "year", value = "y",
                  chg_2011:chg_2019)
df_long <- df_long %>%
  mutate(x = as.numeric(substr(year,5,8)))
df_long2 <- gather(battleground_df, key = "year", value = "n",
                  static_2011:static_2024)
df_long2 <- df_long2 %>%
  mutate(x = as.numeric(substr(year,8,11)))

merged_df <- merge(df_long[c("y", "x", "county_fips", "state_po")],
                  df_long2[c("n", "x", "county_fips", "state_po")],
                  by = c("x", "county_fips", "state_po"), all = TRUE)
merged_df <- unique(merged_df)
merged_df <- dplyr::select(merged_df, county_fips, state_po, x, n, y)
```

The above data shows the two predictors, x (year) and n (static population), and the response variable, y, the change of valence. Here it is in table format:

County FIP	State	Year	Population	Valence Change
4001	AZ	2011	48121	198
4003	AZ	2011	94084	-651
4005	AZ	2011	93356	-366
4007	AZ	2011	36903	7
4009	AZ	2011	22540	1
4011	AZ	2011	5791	49
4012	AZ	2011	10900	-19
4013	AZ	2011	3006087	11413

Use the factor to divide the data by population into color-coded sections.

```
# To ascertain the rural vs urban division, create another field with population grouping info.
# "n" is the field of people who remain, the static population.
labels <- c("tiny", "small", "medium", "large", "very large", "enormous")
merged_df$population_group <- cut(merged_df$n,
                                c(0, 10000, 50000, 100000, 500000, 1000000, Inf),
```

```

labels = labels)
merged_df$population_group <- factor(merged_df$population_group,
                                     levels = c("tiny", "small", "medium",
                                                "large", "very large", "enormous"))

```

In the following scatterplot, the valence change for each county is plotted in the color of its population group factor. There is a wide contrast. Tiny counties are smaller than 10000 in population, small are less than 50000, medium less than 100000, green less than 500000, blue less than a million, and violet include the largest counties of the United States.

Here are the largest and smallest of them:

```

df_sorted_by_population <- battleground_df %>%
  arrange(desc(static_2019)) %>%
  distinct(county_fips, .keep_all = TRUE)

kable(df_sorted_by_population[1:10, c(1, 2, 3, 21)],
      col.names = c("FIPS", "County Name",
                    "State", "Individuals (2019)"))

```

FIPS	County Name	State	Individuals (2019)
48201	HARRIS	TX	3512880
4013	MARICOPA	AZ	3347906
12086	MIAMI-DADE	FL	2045647
48113	DALLAS	TX	1940761
32003	CLARK	NV	1723005
48439	TARRANT	TX	1633772
48029	BEXAR	TX	1524527
12011	BROWARD	FL	1462390
26163	WAYNE	MI	1296621
12099	PALM BEACH	FL	1115251

```

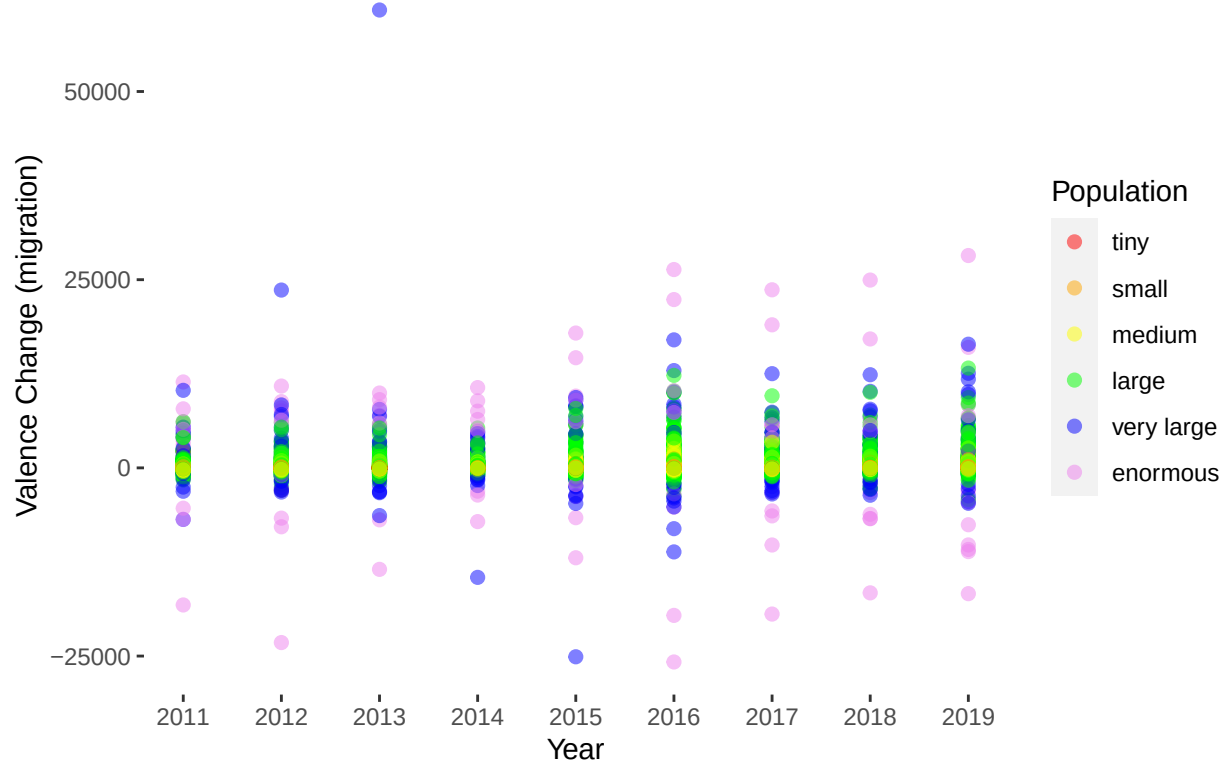
kable(tail(df_sorted_by_population,10)[1:10, c(1, 2, 3, 21)],
      col.names = c("FIPS", "County Name",
                    "State", "Individuals (2019)"))

```

	FIPS	County Name	State	Individuals (2019)
1218	48443	TERRELL	TX	550
1219	31171	THOMAS	NE	544
1220	48033	BORDEN	TX	492
1221	31115	LOUP	NE	482
1222	31009	BLAINE	NE	391
1223	31005	ARTHUR	NE	383
1224	31117	MCPHERSON	NE	379
1225	48261	KENEDY	TX	223
1226	48269	KING	TX	197
1227	48301	LOVING	TX	67

Interestingly, Texas has both the largest and smallest counties in our dataset, and their relative sizes is a factor of 50000! In the following scatterplot each county has a data point in the graphs.

Change in Political Valence in Battleground States



Aside from the principle that most of the valence change occurs in larger counties and a general bias to positive (Democratic) change, not much else can be determined from this graph. Later, each of the states of our questions will be drilled into, and other trends will become apparent. For now, though, the quantity of the data obscures it somewhat.

Until now, I have been merely merging, summarizing, or rearranging the original data. The next step is to predict how the valence will change in future years. I already used exponential growth methods to calculate future populations. Now I will use these values to generate new future valence change using two predictors: the year (x) and the population (n).

First, I will try a clustered version of the k-nearest neighbors algorithm, separating my data into the `population_group` factors from before.

```
known_data_df <-
  merged_df[merged_df$x >= 2011 & merged_df$x <= 2019, ]

prediction_results_df_list <- vector("list", 6)

for (label in 1:length(labels)) {
  factor_name <- labels[label]
  predictors <-
    known_data_df[known_data_df$population_group == factor_name, c("x", "n")]
  responses <-
    known_data_df$y[known_data_df$population_group == factor_name]

  # Split the data into training and testing sets
  set.seed(123)
  trainIndex <-
    sample(nrow(predictors), size = nrow(predictors) * 0.8)
```

```

predictors_train <- predictors[trainIndex, ]
responses_train <- responses[trainIndex]
predictors_test <- predictors[-trainIndex, ]
responses_test <- responses[-trainIndex]

# Perform KNN regression on the training set
predictions <- FNN::knn.reg(
  train = predictors_train,
  test = predictors_test,
  y = responses_train,
  k = 5
)

prediction_results_df_list[[label]] <- data.frame(
  x = predictors_test[, 1],
  n = predictors_test[, 2],
  actual = responses_test,
  guess = predictions$pred
)

# Evaluate the model
mse <-
  mean((prediction_results_df_list[[label]]$actual -
        prediction_results_df_list[[label]]$guess) ^ 2)
print(paste0("Mean Squared Error: ", mse))
}

```

```

## [1] "Mean Squared Error: 938.541703470032"
## [1] "Mean Squared Error: 35075.8860681115"
## [1] "Mean Squared Error: 249407.993898305"
## [1] "Mean Squared Error: 2754129.92"
## [1] "Mean Squared Error: 76282787.0538775"
## [1] "Mean Squared Error: 99625871.2556522"

```

These are dreadful results. Clearly, even accounting for sociological behavioral changes in the different settlement sizes, KNN is not a great match for the data.

I discounted simple linear regression because of the scatter plots and exponential factor for population growth, but it is worth seeing how it compares to knn.

```

lm_prediction_results_df_list <- vector("list", 6)

for (label in 1:length(labels)) {
  factor_name <- labels[label]
  factor_subset_df <- known_data_df[known_data_df$population_group == factor_name,]

  # Split the data into training and testing sets
  set.seed(123)
  trainIndex <-
    sample(nrow(factor_subset_df), size = nrow(factor_subset_df) * 0.8)

  train_df <- factor_subset_df[trainIndex,]
  test_df <- factor_subset_df[-trainIndex,]

  linear_model <- lm(y ~ x + n, train_df)
}

```

```

# Make predictions on the testing set
predictions <- predict(linear_model, newdata = test_df)

# Calculate the MSE
mse <- mean((test_df$y - predictions)^2)

# View the model coefficients and MSE
summary(linear_model)
cat("MSE:", mse, "\n")

lm_prediction_results_df_list[[label]] <- data.frame(
  x = test_df$x,
  n = test_df$n,
  actual = test_df$y,
  guess = predictions
)
# Evaluate the model
}

```

```

## MSE: 809.9644
## MSE: 27224.95
## MSE: 210250.3
## MSE: 2432016
## MSE: 95007951
## MSE: 137844050

```

These are even worse results. I also tried neural network (nnet) and a few others but was unable to find a good algorithm that matches the data. Instead, a simpler traditional linear method on each row actually works better and produces credible results.

```

# estimate 2020+ data from 2011-2019 data.

simple_predicted_df <- data.frame()
future_valence_changes <- array(0, dim = c(nrow(battleground_df), 5))

# Loop through each row of the data frame
for(fip_ix in 1:nrow(battleground_df)) {

  # Get the data for the current row
  current_row <- battleground_df[fip_ix,]

  # Extract the population data for 2011-2019
  y_linear <- as.numeric(current_row[4:12])
  n_linear <- as.numeric(current_row[13:21])
  # Create a vector with the years for 2020-2024
  x_linear <- c(2011:2019)

  x_new <- c(2020:2024)
  n_new <- as.numeric(current_row[22:26])

  # Fit a linear regression model to the data
  simple_model <- lm(y_linear ~ x_linear + n_linear)

  future_valence_changes[fip_ix, ] <-
    predict(simple_model, newdata = data.frame(x_linear = x_new, n_linear = n_new))
}

```

```

}

battleground_df[, c("chg_2020", "chg_2021", "chg_2022",
                    "chg_2023", "chg_2024")] <- round(future_valence_changes,0)
battleground_df<-select(battleground_df, county_fips, county_name, state_po,
                        chg_2011, chg_2012, chg_2013, chg_2014, chg_2015,
                        chg_2016, chg_2017, chg_2018, chg_2019, chg_2020,
                        chg_2021, chg_2022, chg_2023, chg_2024,
                        static_2011, static_2012, static_2013, static_2014,
                        static_2015, static_2016, static_2017, static_2018,
                        static_2019, static_2020, static_2021, static_2022,
                        static_2023, static_2024)

# save the data, so we can use it as a short cut
export(battleground_df, "data/battleground.csv")

# regather our numeric data to include the predicted values
df_long <- gather(battleground_df, key = "year", value = "y",
                  chg_2011:chg_2024)
df_long <- df_long %>%
  mutate(x = as.numeric(substr(year,5,8)))
df_long2 <- gather(battleground_df, key = "year", value = "n",
                  static_2011:static_2024)
df_long2 <- df_long2 %>%
  mutate(x = as.numeric(substr(year,8,11)))

merged_df <- merge(df_long[c("y", "x", "county_fips", "state_po")],
                  df_long2[c("n", "x", "county_fips", "state_po")],
                  by = c("x", "county_fips", "state_po"), all = TRUE)
merged_df <- unique(merged_df)
merged_df <- dplyr::select(merged_df, county_fips, state_po, x, n, y)
# need to reconstruct factor too.
merged_df$population_group <- cut(merged_df$n,
                                c(0, 10000, 50000, 100000, 500000, 1000000, Inf),
                                labels = labels)
merged_df$population_group <- factor(merged_df$population_group,
                                   levels = c("tiny", "small", "medium",
                                              "large", "very large", "enormous"))

# Save this intermediate data too.
rio::export(merged_df, "data/battlemath.csv")

# we can now use merged_df for plotting through 2024.
# recall the previous demo data for the valence calendar.
# use the same subset to show the battleground estimates.
demo_state_test <- battleground_df$state_po %in% demo_states_subset
demo_county_test <- battleground_df$county_name %in% demo_counties_subset

demo_battleground_df<-battleground_df[demo_state_test & demo_county_test,]
demo_battleground_df <- demo_battleground_df %>%
  mutate(location = paste(demo_battleground_df$county_name,
                         demo_battleground_df$state_po, sep=" ", ))

```

```
kable(demo_battleground_df[,c(32, 5, 7, 9, 11, 13, 15, 17)],
      col.names = c("Location", "2012", "2014", "2016", "2018",
                    "2020", "2022", "2024"))
```

	Location	2012	2014	2016	2018	2020	2022	2024
116	MARICOPA, AZ	8804	8901	26345	24950	28933	34108	39231
431	COBB, GA	2040	1449	543	663	315	-48	-412
454	FAYETTE, GA	479	100	1336	765	730	865	1000
473	HENRY, GA	1298	1437	4106	3144	3713	4405	5090
501	MONTGOMERY, GA	-21	-2	10	6	23	34	45
505	NEWTON, GA	445	359	1014	900	1132	1319	1505
520	ROCKDALE, GA	403	472	861	632	634	714	794
2075	DELAWARE, OH	1830	919	3472	1977	2385	2667	2917
2078	FAYETTE, OH	46	53	3	53	137	158	179
2085	HAMILTON, OH	-1755	-902	-2213	-1516	-1819	-1930	-2041
2089	HENRY, OH	-37	-13	8	49	-7	9	26
2111	MONTGOMERY, OH	-967	-773	-1075	-496	-583	-505	-427
2270	CHESTER, PA	1218	29	2379	1027	2023	2393	2763
2278	DELAWARE, PA	430	541	1383	1165	1292	1536	1781
2281	FAYETTE, PA	-7	39	-104	-37	-13	-2	9
2301	MONTGOMERY, PA	1428	859	4417	2942	3968	4784	5600
2576	COLLIN, TX	7090	4000	17005	10168	13581	15257	16616
2594	DENTON, TX	5410	4172	12927	12372	11245	12712	13943
2608	FAYETTE, TX	16	28	146	55	166	209	252
2630	HAMILTON, TX	4	-14	29	-3	-1	2	4
2703	MONTGOMERY, TX	5060	3065	9945	6024	7164	7678	7990
2709	NEWTON, TX	-41	-13	-29	-7	-3	-7	-11
2753	TARRANT, TX	6329	4974	10247	5718	6857	7083	7272

This clearly describes Frey’s nightmare of impending blue rule. Though it is unrealistic to assume all conditions remain the same in politics, this is by far the model that gives the most realistic outcome.

At this point, a summation of the effects of migration is in order. After all, movement from urban centers to suburbs or rural areas in the same state might effect the composition of the House of Representatives, but by itself it does not impact presidential or senatorial outcomes.

```
# summarize net valence
# Create a subset of the data with the states and valence changes
df_subset <- select(battleground_df, state_po, chg_2011, chg_2012, chg_2013,
                    chg_2014, chg_2015, chg_2016, chg_2017, chg_2018, chg_2019,
                    chg_2020, chg_2021, chg_2022, chg_2023, chg_2024)

# Sum the votes by state
df_sum <- df_subset %>%
  group_by(state_po) %>%
  summarize(total_chg_2011 = sum(chg_2011),
            total_chg_2012 = sum(chg_2012),
            total_chg_2013 = sum(chg_2013),
            total_chg_2014 = sum(chg_2014),
            total_chg_2015 = sum(chg_2015),
            total_chg_2016 = sum(chg_2016),
            total_chg_2017 = sum(chg_2017),
            total_chg_2018 = sum(chg_2018),
```

```

total_chg_2019 = sum(chg_2019),
total_chg_2020 = sum(chg_2020),
total_chg_2021 = sum(chg_2021),
total_chg_2022 = sum(chg_2022),
total_chg_2023 = sum(chg_2023),
total_chg_2024 = sum(chg_2024)

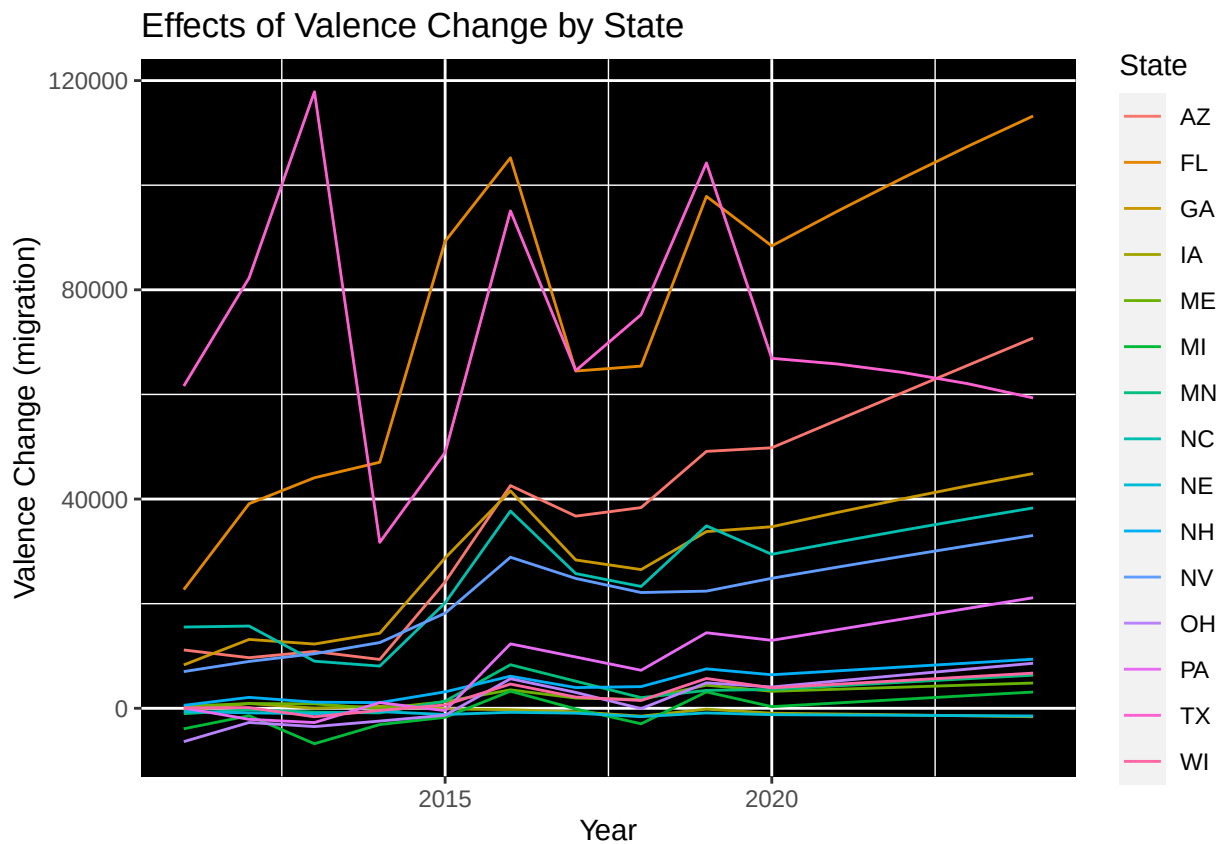
)

df_long_sum <- gather(df_sum, key = "year_key", value = "Valence_Change",
                      total_chg_2011:total_chg_2024)
df_long_sum <- df_long_sum %>%
  mutate(year = as.numeric(substr(year_key,11,15)))

```

Now there is a simple x-y relationship, where x is the year and y the annual change in political valence caused by migration.

Any positive changes correspond to a theoretically more Democratic electorate, while negative changes reflect a more favorable climate for Republicans.



In tabular form, the yearly state changes in valence are:

State	2012	2014	2016	2018	2020	2022	2024
AZ	9671	9336	42570	38367	49799	60325	70781
FL	39108	47036	105238	65417	88394	101341	113223
GA	13160	14351	41557	26514	34700	40031	44874
IA	920	365	-274	-1510	-936	-1262	-1615
ME	903	-59	3524	1671	3245	4039	4829

State	2012	2014	2016	2018	2020	2022	2024
MI	-1421	-3108	3274	-2967	311	1718	3104
MN	-287	-873	8326	2032	3589	4980	6325
NC	15723	8063	37699	23286	29438	34016	38300
NE	-870	-638	-765	-1600	-1222	-1338	-1469
NH	2083	1082	6126	4140	6407	7895	9381
NV	8956	12564	28879	22128	24850	29049	33032
OH	-2715	-2442	5655	-70	4069	6359	8604
PA	-2117	1077	12318	7263	12988	17057	21131
TX	82285	31724	95100	75255	66904	64206	59335
WI	120	-408	4634	1487	3915	5302	6721

The Story of the Data

This project's analysis largely corroborates Frey's findings. An ongoing demographic shift to southern urban centers is indeed in progress, and many new citizens originate from northern metropolitan areas. This alone, of course, is not enough to justify a headline like Thompson's "American Migration Patterns Should Terrify the GOP". After all, another reason citizens in Democratic northern states might move to the south is to escape Democratic policies and embrace Republican ideals like lower income taxes. State taxes in New Jersey on \$75,000 income is 6.5%, whereas Florida and Texas have no state income tax at all.

Nevertheless, when these migration results are also bolstered by observed narrowing margins in many states (with Georgia and Arizona actually flipping to Democratic), responsible party leaders should at least take pause and reflect whether their strategies and issue alignment alienates core groups of the public that deserve more consideration.

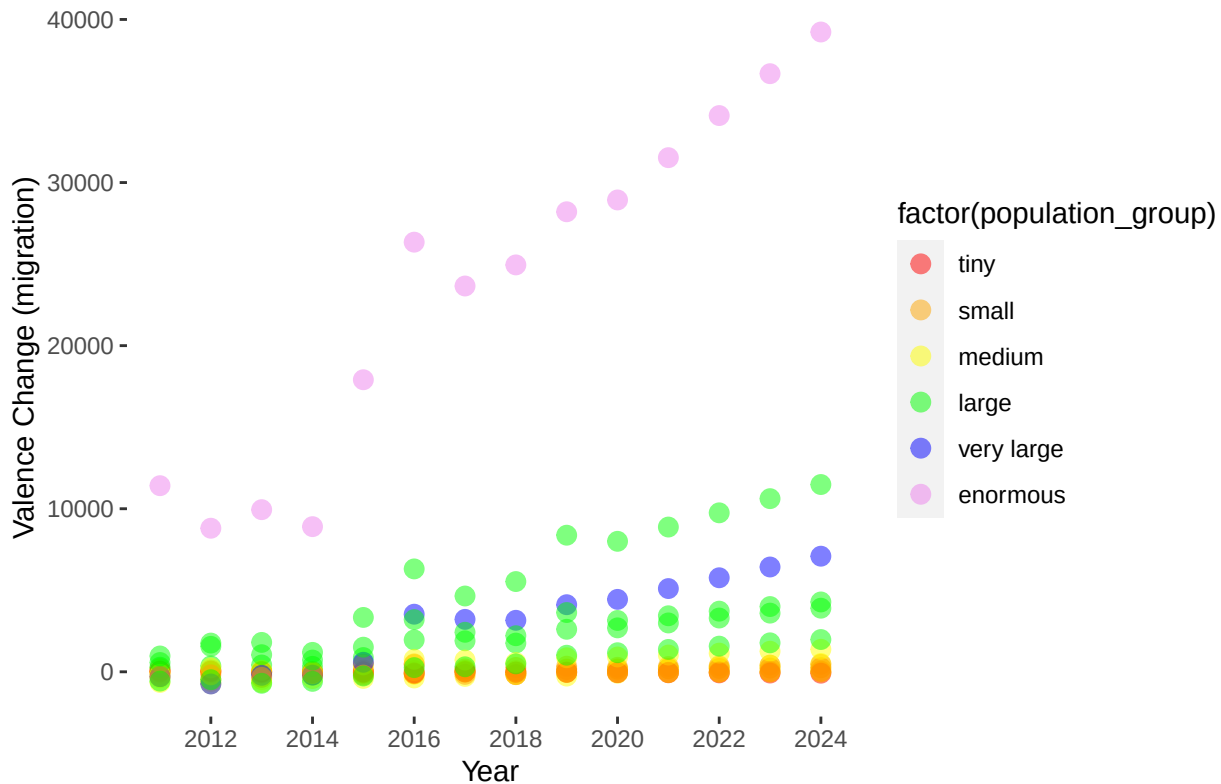
During his 17-term career, the late Massachusetts Representative "Tip" O'Neill Jr. constantly championed the phrase "All politics is local", and to some extent, even in these divided times, he would still be correct. Competent administration generally bests ideology. Good government makes a location more desirable to live, attracting more residents, jobs, and political power.

Earlier a number of questions were raised about various locations undergoing sweeping changes, some due to migration. Each of the stories of these places is different, however, so I will drill down into each and use R and our data to glean more understanding.

Arizona

The last presidential election in Arizona was won by a difference of 0.3%, a little less than 11,000 votes. Our analysis clearly shows the largest counties of Arizona trending Democratic led by Maricopa County, home of Phoenix, America's fifth largest city. If this continues as projected, Arizona will become more hostile towards the GOP. Its 11 electoral votes, vital for the win, are definitely trending blue.

Change in Political Valence in Arizona



In tabular form, the yearly county changes (for counties larger than 100,000 IRS individuals) in valence are:

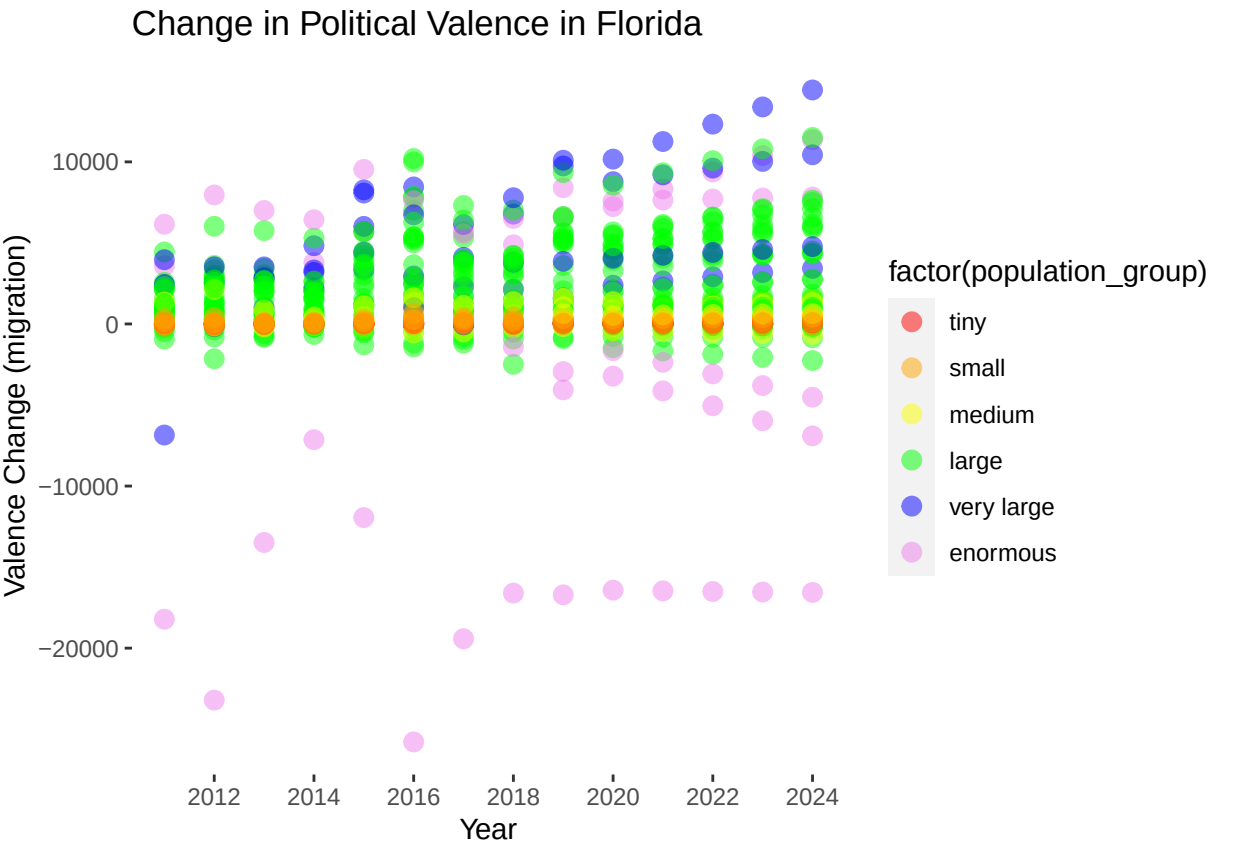
County	Change 2012	Change 2014	Change 2016	Change 2018	Change 2020	Change 2022	Change 2024
MARICOPA	8804	8901	26345	24950	28933	34108	39231
MOHAVE	364	334	1956	1762	2695	3298	3901
PIMA	-746	-205	3533	3158	4442	5766	7089
PINAL	1762	1191	6310	5535	8002	9748	11484
YAVAPAI	1561	721	3202	2213	3149	3720	4284
YUMA	-482	-586	258	487	1171	1575	1979

While Maricopa County is the most important of the state, its other large counties, like Pima and Pinal are also receiving more migrants from Democratic counties. Altogether, Arizona is an area where both parties should concentrate their efforts if they are to be victorious.

Florida

Florida has not elected a Democratic president since 2012, nor a Democratic senator. Media pundits have begun to reclassify Florida as a red state rather than a swing state, and it is a fair assessment. After all, the last Democratic candidate for governor was a former Republican governor, Charlie Crist.

As can be seen in the graph below, Florida is a divided, heterogeneous beast with 30 electoral votes in 2024.



In tabular form, the yearly county changes (for counties larger than 250,000 IRS individuals) in valence are:

County	Change 2012	Change 2014	Change 2016	Change 2018	Change 2020	Change 2022	Change 2024
BREVARD	2487	2111	7816	3725	4581	5308	6001
BROWARD	2996	3791	2505	-285	-1646	-3081	-4520
COLLIER	2998	2044	2844	3083	4544	5211	5894
DUVAL	-180	2249	2960	2130	2391	2914	3428
HILLSBOROUGH	255	4829	8455	4888	7240	9377	11371
LAKE	2866	1550	7022	4228	4948	5605	6147
LEE	6027	5299	7869	6776	8778	9617	10438
MANATEE	3333	1968	5367	3009	3928	4190	4370
MARION	1195	763	3660	2167	3286	3897	4505
MIAMI- DADE	-23208	-7138	-25786	-16605	-16423	-16506	-16563
ORANGE	2581	3178	1004	-1405	-3219	-5045	-6903
OSCEOLA	2807	1660	5390	3418	4143	4306	4305
PALM BEACH	7959	6427	7606	6510	7587	7707	7833
PASCO	3612	2586	10209	7020	8563	10075	11511

County	Change 2012	Change 2014	Change 2016	Change 2018	Change 2020	Change 2022	Change 2024
PINELLAS	3529	3331	6732	3825	4054	4416	4775
POLK	2334	2647	9998	7789	10172	12330	14438
SARASOTA	2578	1668	5277	4053	5682	6593	7495
SEMINOLE	1706	918	2839	1078	1054	1045	1025
VOLUSIA	2219	2002	6273	3842	4826	5712	6571

If Florida remains a swing state, the outcome will be largely defined by the bellwether counties of the I-4 Corridor. Its seven counties are:

Hillsborough County (pop 2020: 1,487,764): This county is located on the west coast of Florida and includes the city of Tampa, which is the largest city in the region.

Pinellas County (974,996): This county is also located on the west coast of Florida and includes the cities of Clearwater and St. Petersburg.

Polk County (724,777): This county is located in the central part of Florida and includes the cities of Lakeland and Winter Haven.

Osceola County (398,439): This county is located in the central part of Florida and includes the city of Kissimmee, which is part of the greater Orlando metropolitan area.

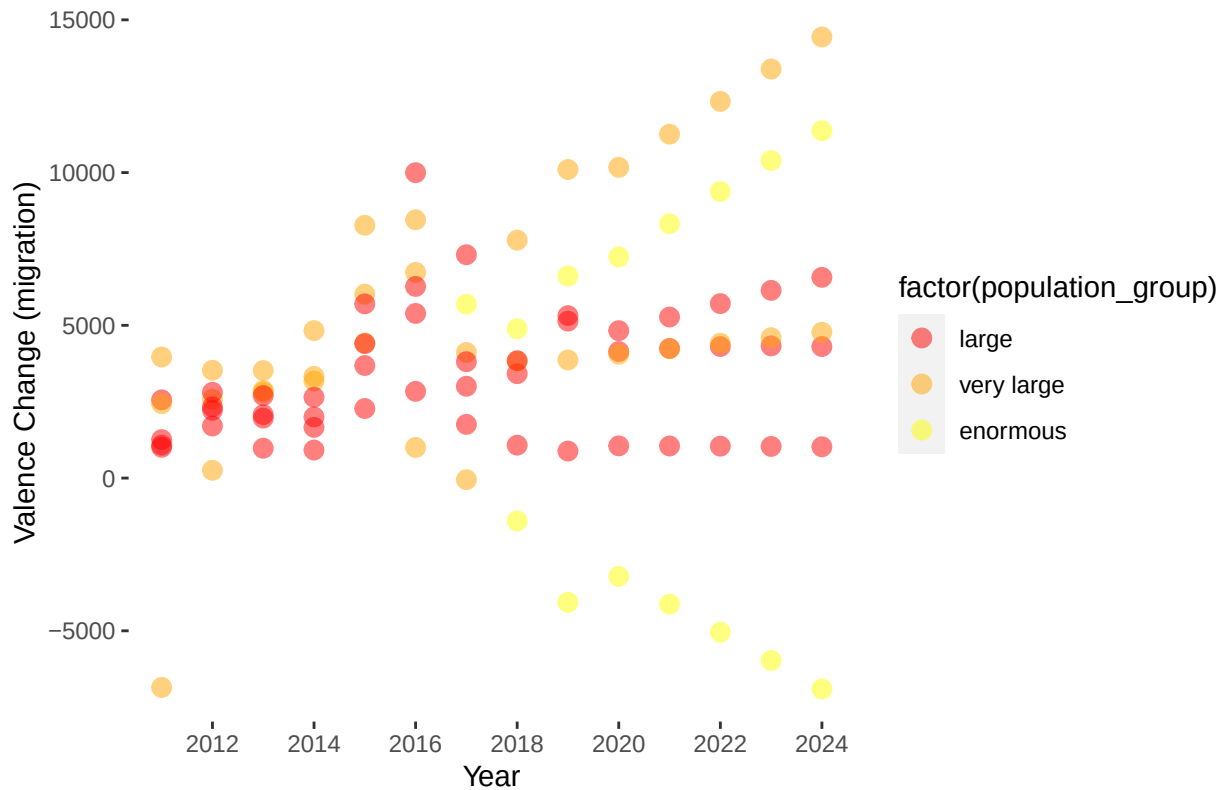
Orange County (1,393,452): This county is located in the central part of Florida and includes the city of Orlando, which is the largest city in the region.

Seminole County (474,081): This county is located in the central part of Florida and includes the city of Sanford, which is part of the greater Orlando metropolitan area.

Volusia County (547,538): This county is located on the east coast of Florida and includes the cities of Daytona Beach and New Smyrna Beach.

A total of 6,001,047 of the total of 21,944,577 Florida citizens, 27 percent. Here is how the change in political valence affects these counties.

Change in Political Valence in Florida



In tabular form, the yearly county changes in valence are:

County	Change 2012	Change 2014	Change 2016	Change 2018	Change 2020	Change 2022	Change 2024
HILLSBOROUGH	255	4829	8455	4888	7240	9377	11371
ORANGE	2581	3178	1004	-1405	-3219	-5045	-6903
OSCEOLA	2807	1660	5390	3418	4143	4306	4305
PINELLAS	3529	3331	6732	3825	4054	4416	4775
POLK	2334	2647	9998	7789	10172	12330	14438
SEMINOLE	1706	918	2839	1078	1054	1045	1025
VOLUSIA	2219	2002	6273	3842	4826	5712	6571

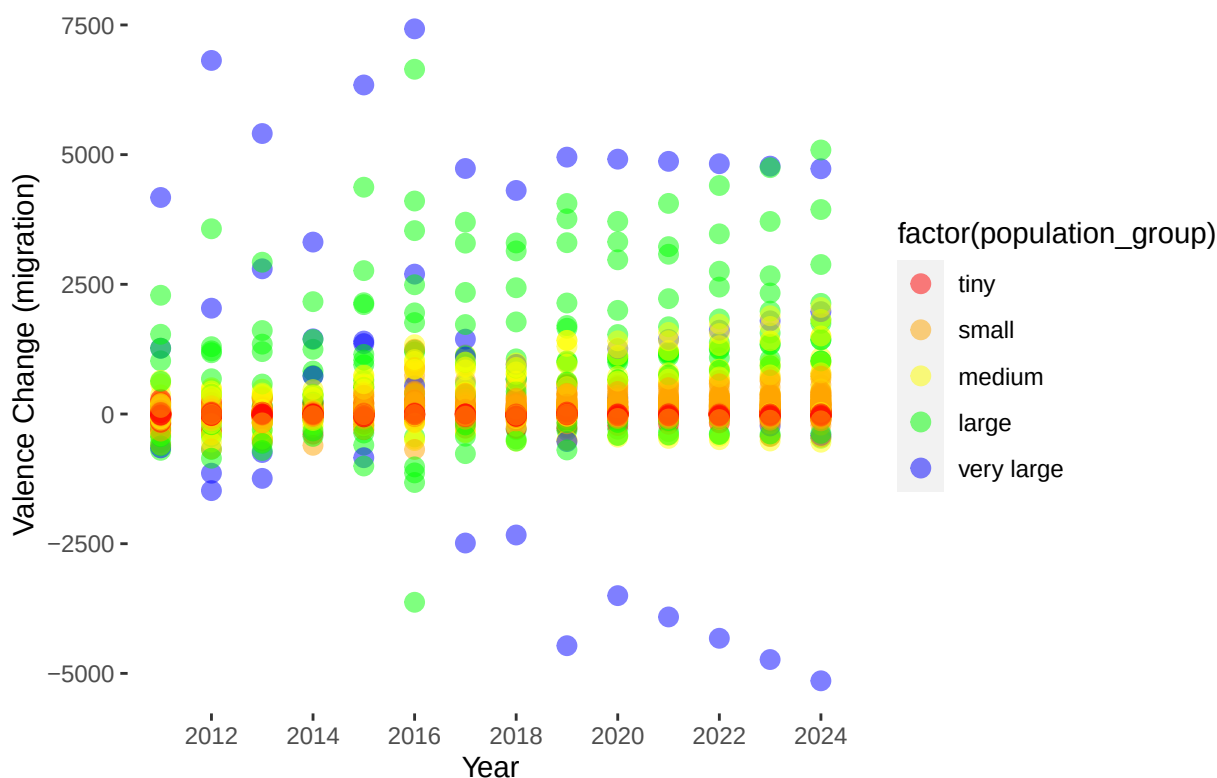
The “enormous” county trending negative is Orange County, a solid Democratic stronghold, whereas Polk County usually votes Republican. In fact, Polk has not voted Democratic since Jimmy Carter won in 1976. In other words, almost anyone who moves to Polk county will swing its valence slightly positive. Likewise, emigrants from Orange County who relocate to other counties will tend to swing Orange County negative. This is an expected outcome of the data.

The general trend across these seven counties is a moderate trend towards positive Democratic outcomes. This does not mean Florida will suddenly turn Democratic in 2024, but there is an underlying migratory trend that offsets its current red aspect.

Georgia

The last presidential election of this state of 16 electoral votes was won by 0.23%, a difference of 11,779 votes, the narrowest of victories in a hotly contested election. Therefore, any change to the state's political climate is significant. Moreover, there is a definite trend to positive valence or a better climate for Democratic candidates.

Change in Political Valence in Georgia



In this graph too, it should be clear that apart from a few outliers (including very large Dekalb County), most of the migration is positive. This confirms the general Democratic trend seen in the Effects of Valence Change by State. Georgia, in fact, is estimated to have a positive valence of over 40000 from 2022 through 2024.

Much of the migration activity is focused on Georgia's largest counties. The following table shows counties with more than 100,000 IRS individual records in the state.

County	Change 2012	Change 2014	Change 2016	Change 2018	Change 2020	Change 2022	Change 2024
BIBB	-682	-126	-441	-301	-122	-28	69
CHATHAM	-382	15	-1135	-481	-293	-381	-468
CHEROKEE	1181	828	3535	2434	2973	3473	3942
CLAYTON	-437	1247	1763	671	1084	1260	1435
COBB	2040	1449	543	663	315	-48	-412
COLUMBIA	1217	743	1950	1063	1127	1085	1021
COWETA	-1	272	1194	680	999	1374	1749
DEKALB	-1477	732	-3629	-2332	-3501	-4323	-5143
FORSYTH	3571	2166	6646	3298	3319	2753	1817
FULTON	-1141	230	2697	956	1261	1617	1974
GWINNETT	6816	3314	7425	4310	4913	4825	4728

County	Change 2012	Change 2014	Change 2016	Change 2018	Change 2020	Change 2022	Change 2024
HALL	680	350	1230	884	1532	1833	2133
HENRY	1298	1437	4106	3144	3713	4405	5090
HOUSTON	76	109	384	434	660	842	1024
MUSCOGEE	-206	-431	-1320	-60	-162	-88	-5
PAULDING	384	227	2492	1775	1998	2447	2882
RICHMOND	-852	-326	-1019	-521	-413	-395	-373

Bibb County (of Macon), Chatham County (of Savannah), Muscogee (of Columbus), and Richmond (of Augusta) are solidly blue. There are also some traditional red counties that are trending blue, such as Columbia and Houston. The rest of the list is composed of Atlanta suburbs and are highlighted by Clayton, Fulton, Henry, and Gwinnett with positive valence and Dekalb and Cobb with negative valence. Needless to say, Dekalb is a Democratic stronghold and many of its emigrants are likely immigrants of Atlanta’s other suburban counties, including Cherokee, Coweta, Forsyth, Hall, and Paulding, which are red counties that are likely to flip Democratic by next decade if these trends hold.

When Derek Thompson warns that one needs to “Just . . . wait” to see the impact of the migration of blue voters to the South, these Atlanta suburban counties are the epicenter of the political earthquake that awaits the GOP (Thompson (2019)).

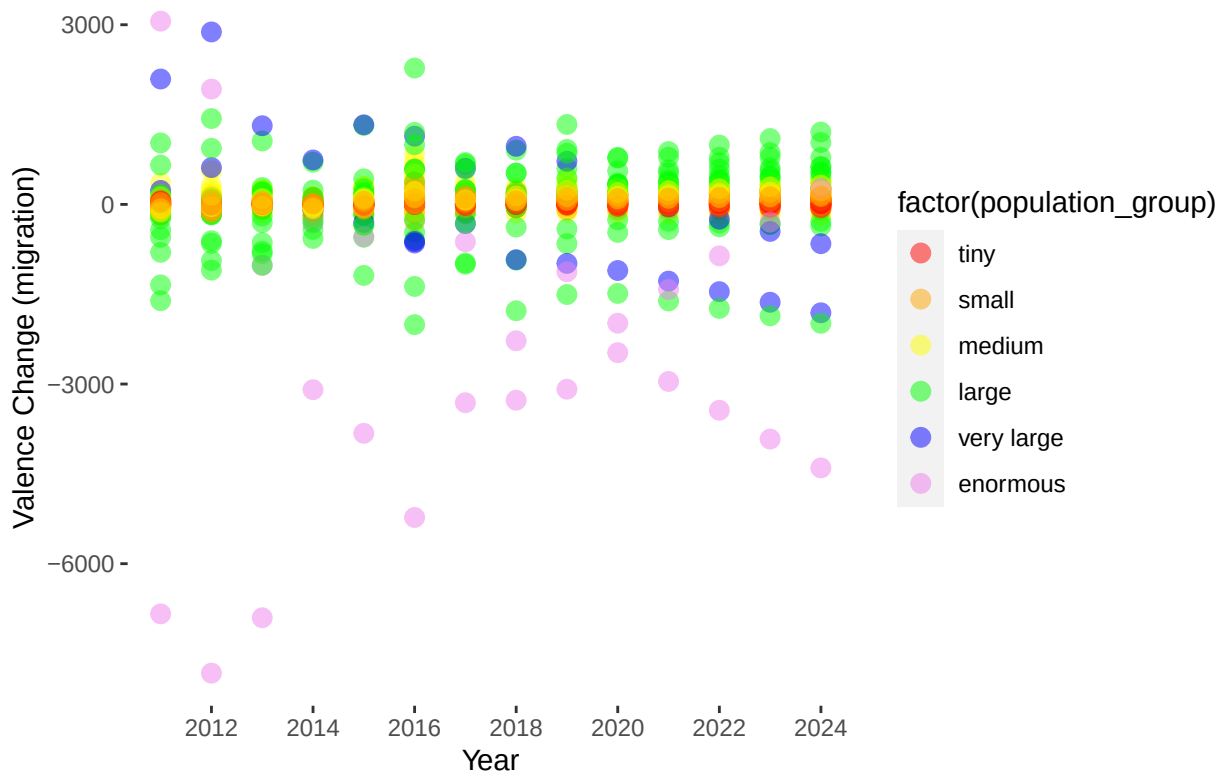
Michigan

On the surface, the data of Michigan appears to portray a far more hospitable state for Republicans, and a drill down into the inflow and outflow data largely corroborates this opinion. However, many of the citizens leaving counties like Kent (-1104 valence change in 2020) are relocating within the state to such counties as Ottawa, Allegan, or Montcalm. As yet, of these destinations, only Ottawa County has shifted left in elections.

Notably, Wayne County, of Michigan's largest city, Detroit, a Democratic stronghold, appears to be ending a trend of bleeding blue voters. Other Democratic strongholds like Kalamazoo and Oakland are likely sourcing many of the changes we see in other states. Maricopa County, for example, is a destination of many counties.

Michigan is close. The last election, won by Biden by only 2.78% or nearly 155,000 votes, was considered a toss-up until nearly the very end of the campaign. As such, it is a state Democrats must defend, as Hillary Clinton learned in 2016, and are opportunities for GOP gains.

Change in Political Valence in Michigan



In tabular form, the yearly county changes (for counties larger than 100,000 IRS individuals) in valence are:

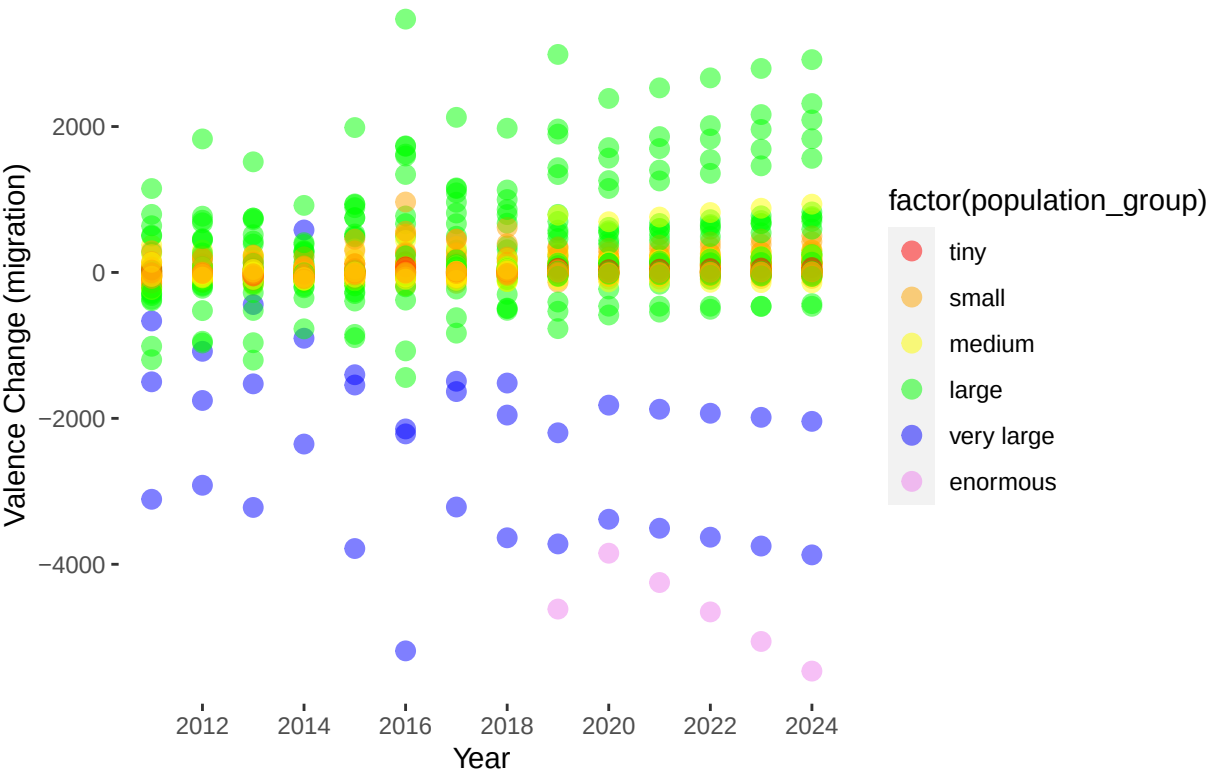
County	Change 2012	Change 2014	Change 2016	Change 2018	Change 2020	Change 2022	Change 2024
BERRIEN	-164	-178	7	110	334	484	633
GENESEE	-937	-416	-464	189	354	694	1034
INGHAM	-1098	-241	-1370	-931	-467	-375	-284
JACKSON	114	-23	583	74	116	191	265
KALAMAZOO	-176	57	-591	-379	-257	-309	-360
KENT	615	-103	-619	-923	-1104	-1457	-1810
LIVINGSTON	1434	705	2277	905	786	787	785
MACOMB	2880	742	1140	970	162	-247	-656
MONROE	12	113	997	523	774	992	1209

County	Change 2012	Change 2014	Change 2016	Change 2018	Change 2020	Change 2022	Change 2024
MUSKEGON	-23	35	375	220	222	289	355
OAKLAND	1926	-313	-645	-2277	-2476	-3438	-4400
OTTAWA	937	239	1202	522	559	593	624
SAGINAW	-606	-295	-251	-55	199	373	547
ST. CLAIR	98	116	592	170	324	422	519
WASHTENAW	-650	-568	-2008	-1776	-1487	-1737	-1985
WAYNE	-7828	-3094	-5228	-3270	-1984	-859	266

Ohio

If there is doubt about Michigan as a welcome state to the GOP, there is none at all with respect to the Buckeye State, which has not elected a Democratic president since 2012. The one bright spot for Democrats is their senator Sherrod Brown, who won in 2018 and will fight a tough reelection campaign in 2024.

Change in Political Valence in Ohio



In tabular form, the yearly county changes (for counties larger than 100,000 IRS individuals) in valence are:

County	Change 2012	Change 2014	Change 2016	Change 2018	Change 2020	Change 2022	Change 2024
BUTLER	-25	50	669	189	440	594	747
CLARK	81	101	98	40	169	202	236
CLERMONT	453	141	494	311	618	723	829
CUYAHOGA	-2917	-2352	-5187	-3637	-3383	-3629	-3871
DELAWARE	1830	919	3472	1977	2385	2667	2917
FAIRFIELD	715	220	1622	999	1259	1547	1833
FRANKLIN	-1081	582	-2145	-1955	-3848	-4653	-5464
GREENE	-523	-348	111	-57	11	141	270
HAMILTON	-1755	-902	-2213	-1516	-1819	-1930	-2041
LAKE	188	299	774	362	546	656	766
LICKING	441	273	1343	675	1151	1358	1564
LORAIN	680	382	1735	1130	1711	2013	2315
LUCAS	-943	-210	-1440	-489	-227	-96	36
MAHONING	-189	-90	-379	-300	-20	-16	-11
MEDINA	771	287	1596	793	571	636	701
MONTGOMERY	-967	-773	-1075	-496	-583	-505	-427
PORTAGE	267	0	227	79	391	492	593

County	Change 2012	Change 2014	Change 2016	Change 2018	Change 2020	Change 2022	Change 2024
STARK	78	-116	-197	52	126	133	140
SUMMIT	-217	-191	-18	-519	-462	-463	-464
TRUMBULL	-175	-3	-42	-97	125	185	245
WARREN	463	408	1729	863	1567	1829	2090

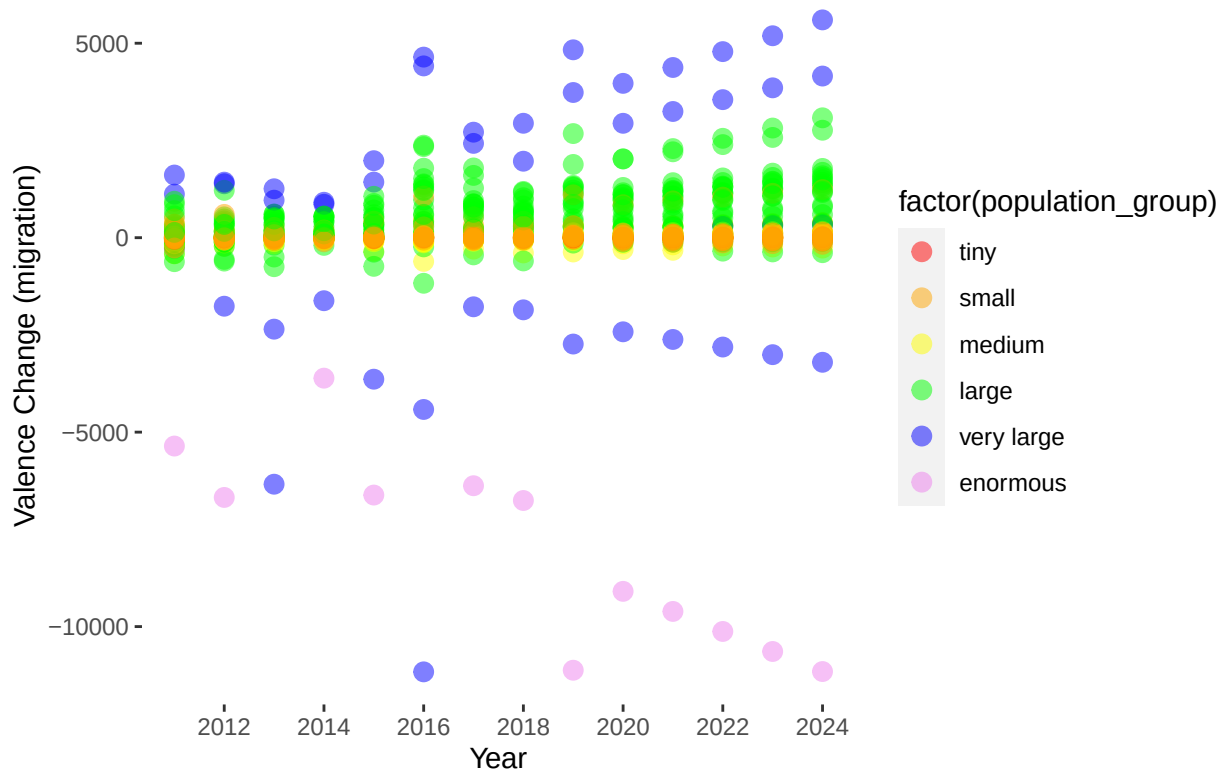
Much of the Democratic exodus in Ohio is focused in three counties: Cuyahoga (Cleveland), Franklin (Columbus), and Hamilton (Cincinnati). These are also the homes of the state's three largest cities and are all Democratic strongholds. Of the positive trending counties, Lorain is a Cleveland suburb (and likely destination of many of its migrants); Delaware, Fairfield, and Licking are similarly suburbs of Columbus; while Butler and Warren are Cincinnati suburbs. In other words, much of the negative valence of the counties of the largest cities is likely relocation within the state.

Ohio is not even close for Democrats though. The last election was won by Republicans by over 8 percent, a total of almost 500,000 voters. There is a slight total positive valence for Ohio, but it seems unlikely that Democrats can make a comeback anytime soon in the state.

Pennsylvania

With 19 electoral votes (down from 20 in the 2020 election), the Keystone State has been an important conquest for campaigns of recent elections. The elections have been close and dramatic. In fact, President Biden won the last election by only 1.17%, or over 80,000 voters. Overall, the state has a general positive valence change, with the highly negative Allegheny (Pittsburgh) and Philadelphia outflows entirely offset by counties like Bucks, Montgomery, Monroe, Chester, and Delaware.

Change in Political Valence in Pennsylvania



In tabular form, the yearly county changes (for counties larger than 100,000 IRS individuals) in valence are:

County	Change 2012	Change 2014	Change 2016	Change 2018	Change 2020	Change 2022	Change 2024
ALLEGHENY	-1762	-1621	-4417	-1855	-2420	-2812	-3204
BEAVER	150	155	297	82	115	122	128
BERKS	166	103	1361	533	830	1125	1420
BUCKS	1387	911	4643	1966	2941	3549	4156
BUTLER	481	287	1237	635	956	1160	1363
CAMBRIA	-116	-45	-73	-49	-4	2	9
CHESTER	1218	29	2379	1027	2023	2393	2763
CUMBERLAND	535	408	967	715	994	1156	1317
DAUPHIN	-224	163	-236	288	467	626	786
DELAWARE	430	541	1383	1165	1292	1536	1781
ERIE	-557	-198	-1172	-598	-117	-109	-95
FRANKLIN	62	79	128	239	277	318	358
LACKAWANNA	-121	82	158	289	538	686	835
LANCASTER	294	-29	419	127	244	284	325
LEBANON	38	65	582	385	233	287	340

County	Change 2012	Change 2014	Change 2016	Change 2018	Change 2020	Change 2022	Change 2024
LEHIGH	-602	555	770	684	1031	1311	1591
LUZERNE	57	471	1304	655	1107	1326	1545
MONROE	-107	540	2341	1190	2028	2555	3082
MONTGOMERY	1428	859	4417	2942	3968	4784	5600
NORTHAMPTON	351	556	1787	968	1155	1418	1680
PHILADELPHIA	-6679	-3612	-11164	-6758	-9094	-10125	-11156
SCHUYLKILL	35	88	347	327	414	505	595
WASHINGTON	382	96	1213	455	718	956	1193
WESTMORELAND	162	179	600	524	808	985	1161
YORK	341	413	1520	812	1105	1295	1486

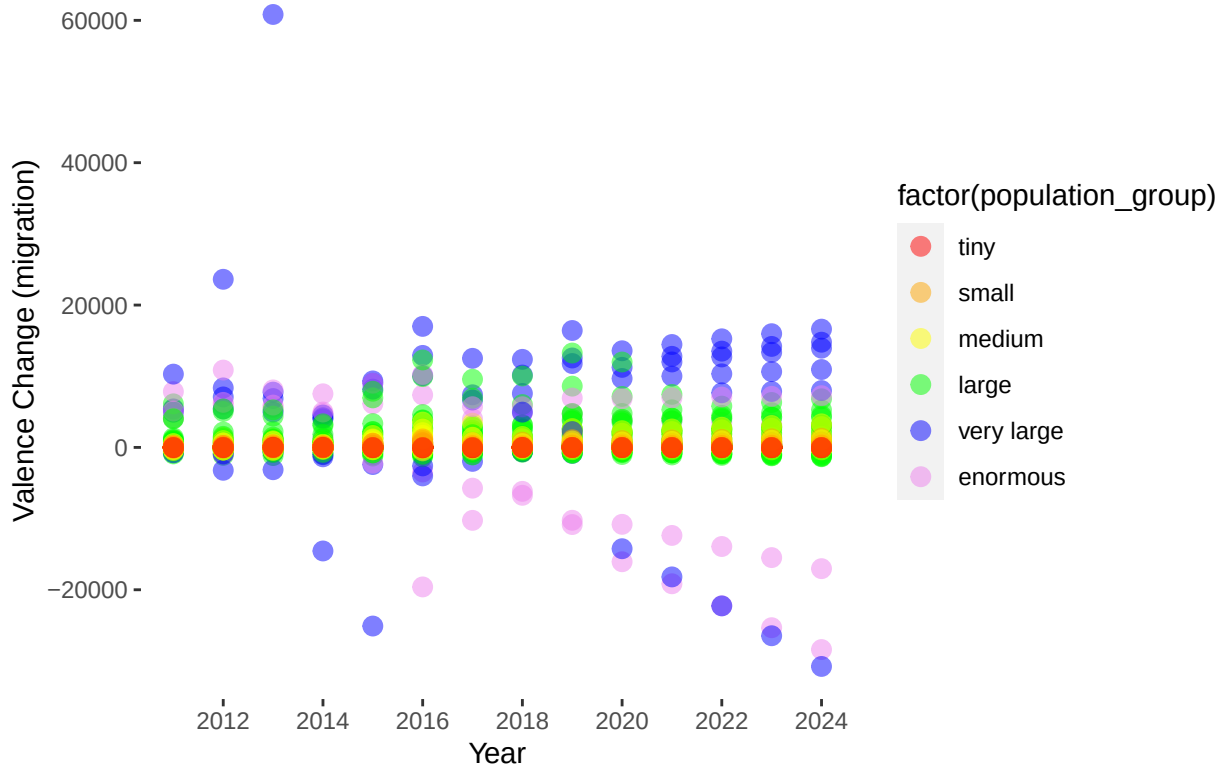
The table shows all the largest counties of the state. My data projects a trend towards a more hospitable climate for Democratic campaigning. Again, however, nothing should be taken for granted.

Texas

Republicans in Texas, which will have 40 electoral votes in 2024 (two more than the 2020 election), won the state by over 5 percentage points, however, the difference was 9 percent in 2016, and nearly 16 percent in 2012. Tarrant County, home of Ft Worth, flipped Democratic in 2020, while other Dallas-Ft Worth suburbs like Denton and Collin Counties also trended left.

It is possible that the GOP can survive losing Georgia and Arizona, but if Texas were to become a blue state, the impact on their party would be catastrophic. A decade ago, few would believe a Democratic victory would be possible, but the demographics of the state are changing, and there is a massive influx of new voters, many of whom originate from blue states.

Change in Political Valence in Texas



In tabular form, the yearly county changes (for counties larger than 250,000 IRS individuals) in valence are:

County	Change 2012	Change 2014	Change 2016	Change 2018	Change 2020	Change 2022	Change 2024
BELL	-1112	608	733	828	1891	2487	3086
BEXAR	6044	4738	7375	4810	3889	3500	3067
BRAZORIA	1282	830	3814	1664	2064	2355	2598
CAMERON	-855	-935	-961	-605	-391	-289	-186
COLLIN	7090	4000	17005	10168	13581	15257	16616
DALLAS	-956	1321	-3453	-6173	-10808	-13922	-17035
DENTON	5410	4172	12927	12372	11245	12712	13943
EL PASO	-3214	-1320	-4003	-611	12	259	509
FORT BEND	8360	4557	10076	7579	9682	10336	10942
GALVESTON	1555	1223	2681	1408	1759	1873	1975
HARRIS	10867	7532	-19600	-6727	-16063	-22247	-28400

County	Change 2012	Change 2014	Change 2016	Change 2018	Change 2020	Change 2022	Change 2024
HIDALGO	-923	-1024	-351	-511	27	187	368
MONTGOMERY	5060	3065	9945	6024	7164	7678	7990
NUECES	165	60	-1305	-330	-635	-854	-1074
TARRANT	6329	4974	10247	5718	6857	7083	7272
TRAVIS	23621	-14548	-2596	4960	-14233	-22268	-30772
WILLIAMSON	5402	3318	12286	10036	11997	13527	14783

Texas is such a large state that the above table only includes counties with over 250,000 people, instead of the 100,000 cut-off of the other states. Some of the trends noted before are present again. Dallas, Harris (home of Houston), and Travis (Austin) have a large net negative valance and are still Democratic strongholds. Much of this is due to relocation within the state.

Bell County (north of Austin) has a positive valance and if trends continue should eventually flip to Democratic. Bexar has consolidated its role as a blue county in recent elections. Brazoria County, a suburb of Houston, has a positive valance but a long to go before flipping. Cameron County, around Brownsville, and neighboring Hidalgo County have been Democratic, but they trended red in 2020, and this is also reflected in the valance data. Tarrant County flipped Democratic in 2020, and nearby Collin and Denton counties have trended blue in recent elections, and although they have not flipped, this could happen very soon. El Paso is already Democratic. Fort Bend, a Houston suburb, flipped blue in 2016 and remained so in the 2020 election, trending even farther left. Galveston remains very red, but Nueces County (Corpus Christi) only leans Republican and may eventually flip. Montgomery County, however, even with a sizable positive valance remains extremely red. Williamson, an Austin suburb, recently flipped Democratic and, as the migration data shows, should continue to trend bluer.

All in all, Texas appears to be almost in play, and that is very bad news for the GOP, who will need to spend funds to defend it. In this way, it is similar to the situation the Democrats have with the Rust Belt states.

Questions for Future Steps

1. I need more migration data. It would be interesting to track past migration data before 2010, which is my current starting point.
2. Rather than exclusively track presidential data, I would like to extend the model in the future to precinct level representative and senate races and extend the idea of “vote” to philosophical questions by tracking the views of the candidates. This requires an objective source for determining issue positional bias. This will make the data more valuable than a simple red/purple/blue correlation.
3. I need to acquire more election results data. This is essential for using past election results to prove the efficacy of the model. At this stage, I am limited to 2000-2020.
4. I need a better method than simple linear regression to predict future results. The algorithms we have learned in class did not generate good predictions. Their mean square errors were very high on prediction of the test data from the training data. There are many more prediction methods in data science, but I require more mastery of these methods.
5. Finally, I would also like to add turnout to my model. This is a very important neglected factor. For most states, turnout data is county based rather than precinct based, so that is an added complication.

Conclusion

The data analysis of this project confirms an ongoing shift of political bias that can impact the results of elections, especially in states with narrow margins of victory. The predictive model generated credible results, but they can be improved upon, which will allow for more accurate forecasting.

In general, the data shows that there is a general move in previously solid red states to purple or, in the case of Arizona, Georgia, and maybe even Texas, to blue in the future. This possibility represents a real threat to the Republican Party and an opportunity for Democrats that must be seized.

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